

# Chapter 3

## **SINGLE-PHASE LIQUID FLOW IN MINICHANNELS AND MICROCHANNELS**

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## 3.1. Introduction

### 3.1.1. Fundamental issues in liquid flow at microscale

Microchannels are used in a variety of devices incorporating single-phase liquid flow.

The early applications involved micromachined devices such as micropumps, microvalves, and microsensors.

This was followed by a thrust in the biological and life sciences with a need for analyzing biological materials, such as proteins, DNA, cells, embryos, and chemical reagents.

The field of micromixers further received attention with developments in microreactors, where two chemical species are mixed prior to introducing them into a reaction chamber. The high flux heat dissipation from high-speed microprocessors provided the impetus for studies on heat transfer in microchannels.

The developments in the microelectromechanical devices naturally require heat removal systems that are equally (LK: fairly) small. Cooling of mirrors employed in high-power laser systems involves cooling systems that cover very small footprints. Advances in biomedical and genetic engineering require controlled fluid transport and its precise thermal control in passages of several micrometer dimensions. A proper understanding of fluid flow and heat transfer in these microscale systems is therefore essential for their design and operation.

In dealing with liquid flows in minichannels and microchannels in the absence of any wall surface effects, such as the electrokinetic or electroosmotic forces that are covered in Chapter 4, the flow is not expected to experience any fundamental changes from the continuum approximation employed in macrofluidic applications.

**LK**

**Air** (in standard conditions): **mean free path is  $\lambda_0 \approx 49$  nm**

**Liquid water:** **mean free path is  $10^5$  times smaller than air!**

$$Kn = \lambda/L_{\text{characteristic}} \quad \text{Very low} \Rightarrow \text{Continuum}$$

Gad-el-Hak (1999) argued that **liquids such as water** should be treated as **continuous media** with the results obtained from classical theory being applicable in channels larger than **1  $\mu\text{m}$** . However, there remain a number of unresolved issues that require further study. The **main areas of current research** are summarized **below**:

- (a) **Experimental validation** of the **laminar** and **turbulent** flow **transport equations** – the laminar flow friction factor and heat transfer equations derived from theoretical considerations are expected to hold in microchannel applications in the absence of any changes in the transport processes or any new physical phenomenon. Although explicit equations and experiments for mass transfer are not covered, the conclusions reached for the momentum and heat transfer are expected to be applicable to mass transport processes as well.
- (b) **Verification** of the **laminar-to-turbulent** flow transition **at microscale** – experimental evidence in this regard needs to be critically evaluated.
- (c) The effect of **large relative roughness** values on the **flow** – **large values** of relative **roughness** are **more commonly** encountered in **microchannels**. Their effect on the **laminar-to-turbulent transition**, **friction factors**, and **heat transfer** needs to be investigated.
- (d) **Verification** of **empirical constants** derived from **macroscale experiments** – a number of constants (such as for the losses due to flow area changes, bends, etc.), whose values are derived from macroscale fluid flow experiments need to be verified for microscale applications.

**This chapter** will be devoted to **answering** the **above questions** on the basis of the evidence available in literature.

### 3.1.2. Need for smaller flow passages

The flow passage dimensions in convective heat transfer applications have been shifting towards smaller dimensions for the following

three main reasons:

- (a) Heat transfer enhancement.
- (b) Increased heat flux dissipation in microelectronic devices.
- (c) Emergence of microscale devices that require cooling.

Employing smaller channel dimensions results in higher heat transfer performance, although it is accompanied by a higher pressure drop per unit length. The higher volumetric heat transfer densities require advanced manufacturing techniques and lead to more complex manifold designs. An optimum balance for each application leads to different channel dimensions.

For example, in the refrigeration industry, the use of microfin tubes of 6–8 mm diameter have replaced the plain tubes of larger diameters. In automotive applications, the passage dimensions for radiators and evaporators have approached a 1 mm threshold as a balance between the pumping power, heat transfer, and cleanliness constraints imposed by the overall system.

Microelectronic devices, which include a variety of applications such as PCs, servers, laser diodes, and RF devices (radio-frequency, LK), are constantly pushing the heat flux density requirements to higher levels.

What seemed to be an impossibly high limit of 200 W/cm<sup>2</sup> of heat dissipation in 1993 now seems to be a feasible target.

The new challenge for the coming decade is on the order of 600–1000 W/cm<sup>2</sup>.

The available temperature differences are becoming smaller, in some cases as low as only a few °C with external copper heat sinks.

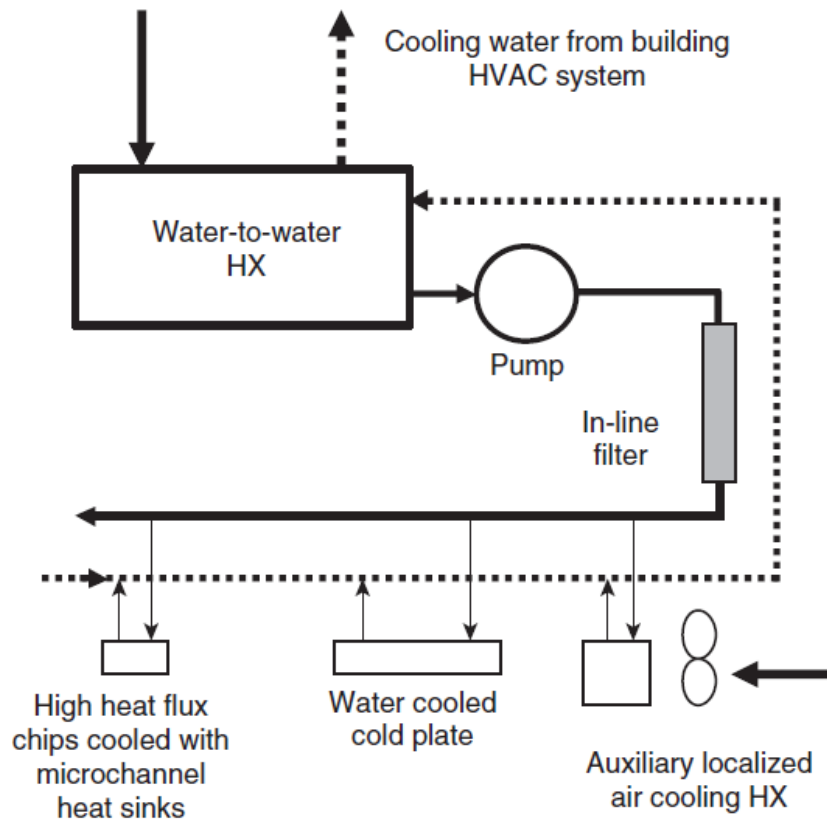
These high levels of heat dissipation require a dramatic reduction in the channel dimensions, matched with suitable coolant loop systems to facilitate the fluid movement away from the heat source.

A cooling system for a microscale device might require cooling channels of a few tens of micrometers as compared to more conventional sized channels with 1–3 mm flow passage dimensions.

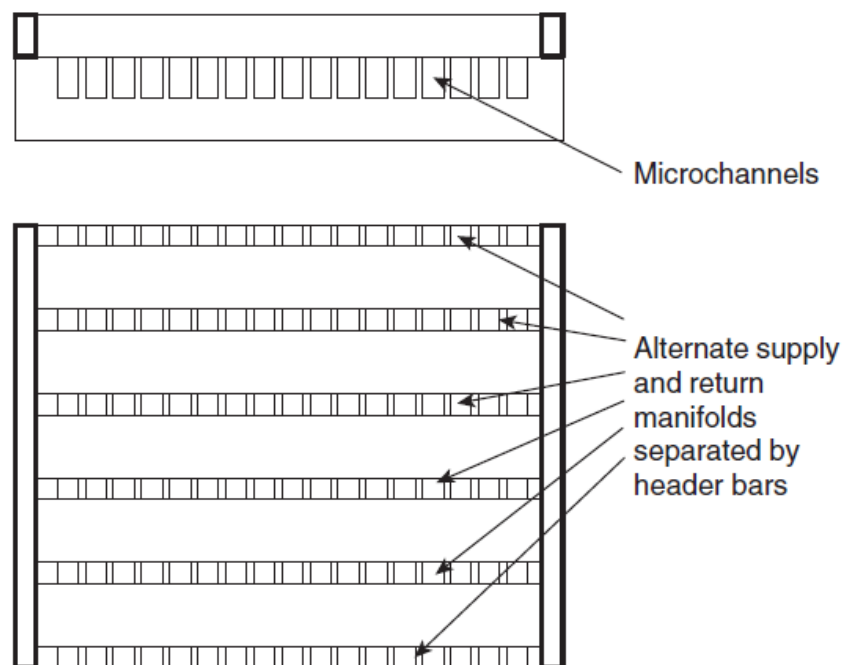
In addition, several such units may be clustered together and a secondary cooling loop may be employed to remove the heat with a conventional cooling system.

Figure 3.1 shows a schematic of a microchannel cooling system configuration for cooling a server application. The combination of the microchannel heat exchangers, mounted directly on the chip or in the heat sink that is bonded to the chip, water cooled cold plates with minichannel or microchannel flow passages, and auxiliary localized cooling systems will be able to address the complex cooling needs of the high send servers. The cooling system is integrated with the building HVAC system as described by Kandlikar (2005).

A schematic of direct liquid cooling of a multichip module or a heat sink is shown in Fig. 3.2. The liquid flows through the cold plates that are attached to a substrate cap. In advanced designs, direct cooling of chips is accomplished by circulating water, a water-antifreeze mixture, oil, or a dielectric fluid, such as FC-72, FC-77 or FC-87 through microchannels that are fabricated on the chip surface. Copper heat sinks with integrated microchannels and minichannels are expected to dominate heat sink applications.



**Fig. 3.1.** Schematic of a cluster of servers with high heat flux chips cooled with microchannel heat sinks, cold plates, and localized air cooling integrated with a secondary chilled water loop from the building HVAC system, Kandlikar (2005).



**Fig. 3.2.** Schematic of a microchannel cooling arrangement on a chip or a heat sink with alternate supply and return manifolds created by header bars, design originally proposed by Tuckerman (1984).

## 3.2. Pressure drop in single-phase liquid flow

### 3.2.1. Basic pressure drop relations

One-dimensional flow of an incompressible fluid in a smooth circular pipe forms the basis for the pressure drop analysis in internal flows. The following equations are readily derived based on the **continuum assumption** for **Newtonian liquid flows** in **minichannels** and **microchannels**.

Considering the equilibrium of a fluid element of **length  $dx$**  in a pipe of **diameter  $D$** , the **force due to pressure difference  $dp$**  is balanced by the frictional force due to **shear stress  $\tau_w$**  at the wall.

$$\left(\frac{\pi}{4}D^2\right)dp = (\pi D dx)\tau_w \quad (3.1)$$

The **pressure gradient** and the **wall shear stress** are thus related by the following equation:

$$\frac{dp}{dx} = \frac{4\tau_w}{D} \quad (3.2)$$

For Newtonian fluids, the **wall shear stress  $\tau_w$**  is expressed in terms of the **velocity gradient** at the wall:

$$\tau_w = \mu \left. \frac{du}{dy} \right|_w \quad (3.3)$$

where  **$\mu$  is the dynamic viscosity**. The **Fanning friction factor  $f$**  is used in heat transfer literature because of its ability to represent the momentum transfer process of fluid flow in a manner consistent with the heat and mass transfer process representations:

$$f = \frac{\tau_w}{(1/2)\rho u_m^2} \quad (3.4)$$

where  **$u_m$  is the mean flow velocity** in the channel.

The frictional pressure drop  $\Delta p$  over a length  $L$  is obtained from Eqs. (3.2) and (3.4) respectively:

$$\frac{dp}{dx} = \frac{4\tau_w}{D} \quad f = \frac{\tau_w}{(1/2)\rho u_m^2}$$

$$\Delta p = \frac{2f\rho u_m^2 L}{D} \quad (3.5)$$

The Fanning friction factor  $f$  in Eq. (3.5) depends on the flow conditions, the channel wall geometry and surface conditions:

- (a) laminar or turbulent flow,
- (b) flow-channel geometry,
- (c) fully developed or developing flow,
- (d) smooth or rough walls.

For non-circular flow channels, the  $D$  in Eq. (3.5) is replaced by the hydraulic diameter  $D_h$  represented by the following equation.

$$D_h = \frac{4A_c}{P_w} \quad (3.6)$$

where  $A_c$  is the flow-channel cross-sectional area and  $P_w$  is the wetted perimeter. For a rectangular channel of sides  $a$  and  $b$ ,  $D_h$  is given by

$$D_h = \frac{4ab}{2(a+b)} = \frac{2ab}{(a+b)} \quad (3.7)$$

### 3.2.2. Fully developed laminar flow

The velocity gradient at the channel wall can be readily calculated from the well-known Hagen–Poiseuille parabolic velocity profile for the fully developed laminar flow in a circular pipe.

Using this velocity profile,  $\tau_w$  and  $f$  are obtained from Eqs. (3.3) and (3.4). The result for friction factor  $f$  is presented in the following form:

$$f = \frac{Po}{Re} \quad (3.8)$$

where  $Po$  is the Poiseuille number, ( $Po = f Re$ ), which depends on the flow-channel geometry.

Table 3.1 gives the  $f Re$  product and the constant Nusselt number in the fully developed laminar flow region for different duct shapes as derived from Kakac et al. (1987).

It can be seen that for a circular pipe,

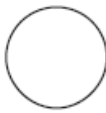
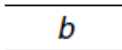
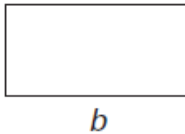
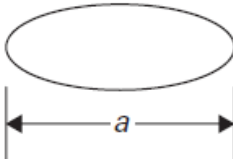
$$Po = f Re = 16 \quad (3.9)$$

Shah and London (1978) provide the following equation for a rectangular channel with short side  $a$  and long side  $b$ , and a channel aspect ratio defined as  $\alpha_c = a/b$ .

$$Po = f Re = 24(1 - 1.3553\alpha_c + 1.9467\alpha_c^2 - 1.7012\alpha_c^3 + 0.9564\alpha_c^4 - 0.2537\alpha_c^5) \quad (3.10)$$

**Table 3.1**

Fanning friction factor and Nusselt number for fully developed laminar flow in ducts, derived from Kakac *et al.* (1987).

| Duct shape                                                                          |                                              | $Nu_H$      | $Nu_T$ | $Po = f Re$ |       |
|-------------------------------------------------------------------------------------|----------------------------------------------|-------------|--------|-------------|-------|
|    | Circular                                     | 4.36        | 3.66   | 16          |       |
|    | Flat channel                                 | 8.24        | 7.54   | 24          |       |
|    | Rectangular,<br>aspect ratio, $b/a =$        | 1           | 3.61   | 2.98        | 14.23 |
|                                                                                     |                                              | 2           | 4.13   | 3.39        | 15.55 |
|                                                                                     |                                              | 3           | 4.79   | 3.96        | 17.09 |
|                                                                                     |                                              | 4           | 5.33   | 4.44        | 18.23 |
|                                                                                     |                                              | 6           | 6.05   | 5.14        | 19.70 |
|                                                                                     |                                              | 8           | 6.49   | 5.60        | 20.58 |
|                                                                                     |                                              | $\infty$    | 8.24   | 7.54        | 24.00 |
|   | Hexagon                                      | 4.00        | 3.34   | 15.05       |       |
|  | Isosceles Triangle,<br>Apex angle $\theta =$ | $10^\circ$  | 2.45   | 1.61        | 12.47 |
|                                                                                     |                                              | $30^\circ$  | 2.91   | 2.26        | 13.07 |
|                                                                                     |                                              | $60^\circ$  | 3.11   | 2.47        | 13.33 |
|                                                                                     |                                              | $90^\circ$  | 2.98   | 2.34        | 13.15 |
|                                                                                     |                                              | $120^\circ$ | 2.68   | 2.00        | 12.74 |
|  | Ellipse,<br>Major/Minor axis $a/b =$         | 1           | 4.36   | 3.66        | 16.00 |
|                                                                                     |                                              | 2           | 4.56   | 3.74        | 16.82 |
|                                                                                     |                                              | 4           | 4.88   | 3.79        | 18.24 |
|                                                                                     |                                              | 8           | 5.09   | 3.72        | 19.15 |
|                                                                                     |                                              | 16          | 5.18   | 3.65        | 19.54 |

$Nu = hD_h/k$ ;  $Re = \rho u_m D_h/\mu$ ;  $Nu_H$  –  $Nu$  under a constant heat flux boundary condition, constant axial heat flux, and uniform circumferential temperature;  $Nu_T$  –  $Nu$  under a constant wall temperature boundary condition;  $f$  – friction factor.

### 3.2.3. Developing laminar flow

As flow enters a duct, the velocity profile begins to develop along its length, ultimately reaching the fully developed Hagen–Poiseuille velocity profile.

Almost all the analyses available in literature consider a uniform velocity condition at the inlet.

The length of the hydrodynamic developing region  $L_h$  is given by the following well-accepted equation:

$$\frac{L_h}{D_h} = 0.05 \text{ Re} \quad (3.11)$$

Since the pressure gradients found in small diameter channels are quite high, the flow lengths are generally kept low.

In many applications, the length of channel in the developing region therefore forms a major portion of the flow length through a microchannel.

To account for the developing region, the pressure drop equations are presented in terms of an apparent friction factor.

Apparent friction factor  $f_{app}$  accounts for the pressure drop due to friction and the developing region effects.

It represents an average value of the friction factor over the flow length between the entrance section and the location under consideration. Thus the pressure drop in a channel of hydraulic diameter  $D_h$  over a length  $x$  from the entrance is expressed as:

$$\Delta p = \frac{2f_{app}\rho u_m^2 L}{D} \quad (3.5)$$

$$\Delta p = \frac{2f_{app}\rho u_m^2 x}{D_h} \quad (3.12)$$

The difference between the apparent friction factor over a length  $x$  and the fully developed friction factor  $f$  is expressed in terms of an incremental pressure defect  $K(x)$ :

$$K(x) = (f_{\text{app}} - f) \frac{4x}{D_h} \quad (3.13)$$

For  $x > L_h$  the incremental pressure defect attains a constant value  $K(\infty)$ , known as *Hagenbach's factor*.

Combining Eqs. (3.12) and (3.13), the pressure drop can be expressed in terms of the incremental pressure drop:

$$\Delta p = \frac{2(f_{\text{app}} \text{Re}) \mu u_m x}{D_h^2} = \frac{2(f \text{Re}) \mu u_m x}{D_h^2} + K(x) \frac{\rho u_m^2}{2} \quad (3.14)$$

For a circular tube, Hornbeck (1964) obtained the axial velocity distribution and pressure drop in a non-dimensional form. He estimated the fully developed region to begin at  $x^+ = 0.0565$ , with a value of  $K(\infty) = 1.28$  for a circular duct.

Chen (1972) proposed the following equation for  $K(\infty)$  for the circular geometry:

$$K(\infty) = 1.20 + \frac{38}{\text{Re}} \quad (3.15)$$

The non-dimensionalized length  $x^+$  given by:

$$x^+ = \frac{x/D_h}{\text{Re}} \quad (3.16)$$

Shah and London (1978) showed that the frictional pressure drop in the developing region of a circular duct obtained by Hornbeck (1964) can be accurately described by the following equation:

$$\frac{\Delta p}{(1/2) \rho u_m^2} = 13.74(x^+)^{1/2} + \frac{1.25 + 64x^+ - 13.74(x^+)^{1/2}}{1 + 0.00021(x^+)^{-2}} \quad (3.17)$$

Rectangular geometries are of particular interest in microfluidics applications. Shah and London (1978) and Kakac et al. (1987) present comprehensive summaries of the available literature. Phillips (1987) reviewed the available information, including that by Curr et al. (1972) and compiled the results for the apparent friction factor in a rectangular duct as shown in Fig. 3.3. It can be seen that fully developed flow is attained at different  $x^+$  values, with low aspect ratio ducts reaching it earlier. The constant in Eq. (3.11) is around 1 ( $x^+ = 10^0$ ) for rectangular ducts as seen from Fig. 3.3.

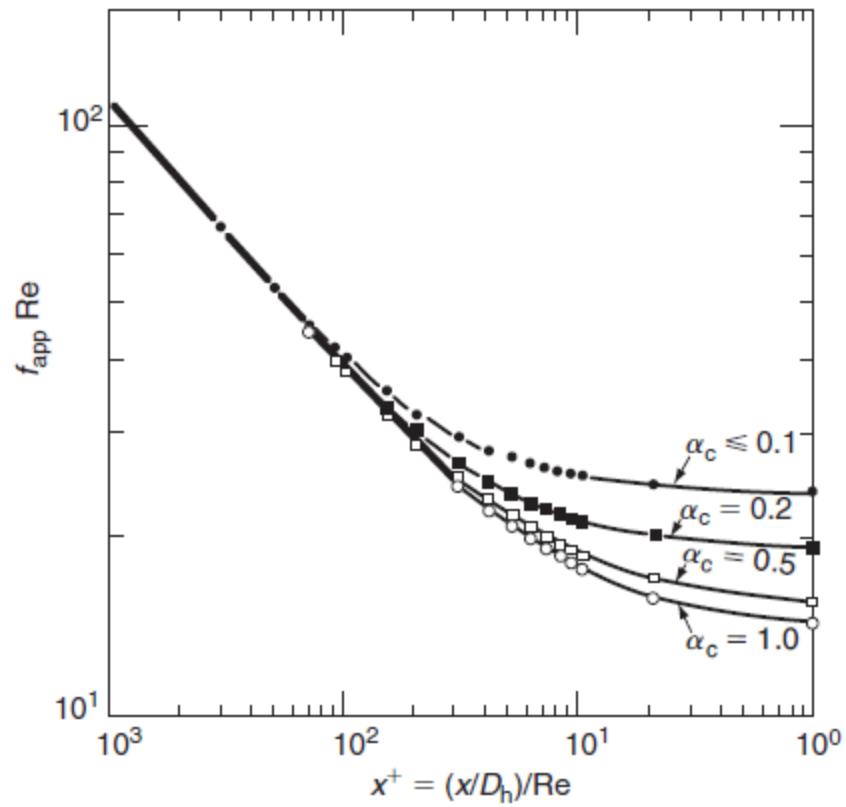
Table 3.2 derived from Phillips (1987) gives the values of the apparent friction factor in a tabular form.

For other channel aspect ratios, a linear interpolation is suggested. Alternatively, use the curve-fit equations provided in Appendix A.

The results for the fully developed friction factors and Hagenbach's factors in trapezoidal channels are reported by Kakac et al. (1987). By considering the rectangular channels as a subset of the trapezoidal geometry, Steinke and Kandlikar (2005a) obtained the following curve-fit equation for the Hagenbach's factor for rectangular channels:

$$K(\infty) = 0.6796 + 1.2197\alpha_c + 3.3089\alpha_c^2 - 9.5921\alpha_c^3 + 8.9089\alpha_c^4 - 2.9959\alpha_c^5 \quad (3.18)$$

The analysis presented in this section assumes a uniform velocity profile at the entrance of the channel. In many microfluidic applications, the channels have the manifold surfaces flush with the two opposing channel surfaces. The effect of such an arrangement was investigated numerically by Gamrat et al. (2004). They showed that the resulting apparent friction factors in the entrance region could be up to 50% lower than the theoretical predictions.



**Fig. 3.3.** Apparent friction factors for rectangular ducts in the developing region for different aspect ratios ( $\alpha = 1/\alpha_c$ ), Phillips (1987).

**Table 3.2**

Laminar flow friction factor in the entrance region of rectangular ducts.

| $x^+ = (x/D_h)/Re$ | $f_{app} Re$     |                  |                  |                                           |
|--------------------|------------------|------------------|------------------|-------------------------------------------|
|                    | $\alpha_c = 1.0$ | $\alpha_c = 0.5$ | $\alpha_c = 0.2$ | $\alpha_c \leq 0.1$<br>$\alpha_c \geq 10$ |
| 0                  | 142.0            | 142.0            | 142.0            | 287.0                                     |
| 0.001              | 111.0            | 111.0            | 111.0            | 112.0                                     |
| 0.003              | 66.0             | 66.0             | 66.1             | 67.5                                      |
| 0.005              | 51.8             | 51.8             | 52.5             | 53.0                                      |
| 0.007              | 44.6             | 44.6             | 45.3             | 46.2                                      |
| 0.009              | 39.9             | 40.0             | 40.6             | 42.1                                      |
| 0.01               | 38.0             | 38.2             | 38.9             | 40.4                                      |
| 0.015              | 32.1             | 32.5             | 33.3             | 35.6                                      |
| 0.02               | 28.6             | 29.1             | 30.2             | 32.4                                      |
| 0.03               | 24.6             | 25.3             | 26.7             | 29.7                                      |
| 0.04               | 22.4             | 23.2             | 24.9             | 28.2                                      |
| 0.05               | 21.0             | 21.8             | 23.7             | 27.4                                      |
| 0.06               | 20.0             | 20.8             | 22.9             | 26.8                                      |
| 0.07               | 19.3             | 20.1             | 22.4             | 26.4                                      |
| 0.08               | 18.7             | 19.6             | 22.0             | 26.1                                      |
| 0.09               | 18.2             | 19.1             | 21.7             | 25.8                                      |
| 0.10               | 17.8             | 18.8             | 21.4             | 25.6                                      |
| 0.20               | 15.8             | 17.0             | 20.1             | 24.7                                      |
| >1.0               | 14.2             | 15.5             | 19.1             | 24.0                                      |

For intermediate values use the curve-fit equations provided in Appendix A at the end of the chapter.

### 3.2.4. Fully developed and developing turbulent flow

A number of correlations with comparable accuracies are available in literature for fully developed turbulent flow in smooth channels. The following equation by Blasius is used extensively:

$$f = 0.0791 \text{Re}^{-0.25} \quad (3.19)$$

A more accurate equation was presented by Phillips (1987) to cover both the developing and fully developed flow regions. He presented the fanning friction factor for a circular tube in terms of the following equation:

$$f_{\text{app}} = A \text{Re}^B \quad (3.20)$$

where

$$A = 0.09290 + \frac{1.01612}{x/D_h} \quad (3.21)$$

$$B = -0.26800 - \frac{0.32930}{x/D_h} \quad (3.22)$$

For rectangular channel geometries,

**Re** is replaced with the laminar-equivalent Reynolds number (Jones, 1976) given by:

$$\text{Re}^* = \frac{\rho u_m D_{le}}{\mu} = \frac{\rho u_m [(2/3) + (11/24)(1/\alpha_c)(2 - 1/\alpha_c)] D_h}{\mu} \quad (3.23)$$

where  $D_{le}$  is the laminar-equivalent diameter given by the term in the square parenthesis in Eq. (3.23).

$$D_{le} = [(2/3) + (11/24)(1/\alpha_c)(2 - 1/\alpha_c)] \quad \text{LK}$$

### 3.3. Total pressure drop in a microchannel heat exchanger

#### 3.3.1. Friction factor

An earnest interest in microchannel flows began with the pioneering work on direct chip cooling with water by Tuckerman and Pease (1981). Recently, a number of investigators including Li et al. (2000), Celata et al. (2002) and Steinke and Kandlikar (2005a) critically evaluated the available literature and presented explanations for the large deviations from the classical theory reported by some of the researchers.

A schematic representation of the pressure drop experiments conducted by the researchers is shown in Fig. 3.4.

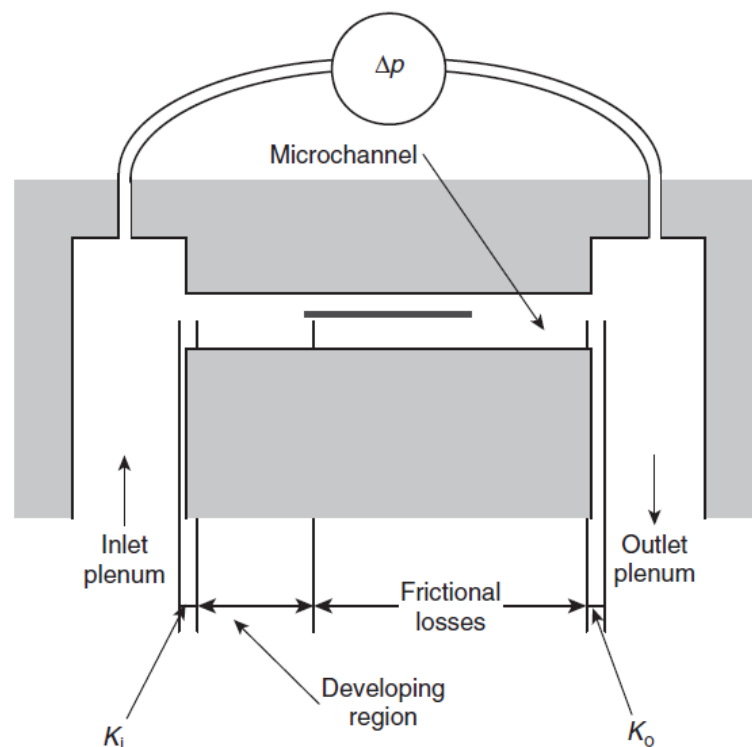


Fig. 3.4. Schematic representation of the experiments employed by researchers for pressure drop measurements in microchannels, Steinke and Kandlikar (2005a).

Since measuring the local pressure along the flow is difficult in microchannels, researchers have generally measured the pressure drop across the inlet and outlet manifolds.

The resulting pressure drop measurement represents the combined effect of the losses in the

bends, entrance and exit losses,

developing region effects, and the core frictional losses.

Thus, the measured pressure drop is the sum of these components (Phillips, 1987):

$$\Delta p = \frac{\rho u_m^2}{2} \left[ (A_c/A_p)^2 (2K_{90}) + (K_c + K_e) + \frac{4f_{app}L}{D_h} \right] \quad (3.24)$$

where

$A_c$  and  $A_p$  are the total channel area and the total plenum cross-sectional area,  $K_{90}$  is the loss coefficient at the 90-degree bends,

$K_c$  and  $K_e$  represent the contraction and expansion loss coefficients due to area changes, and

$f_{app}$  includes the combined effects of frictional losses and the additional losses in the developing flow region.

Phillips (1987) studied these losses and recommends  $K_{90}$  to be approximately 1.2. The contraction and exit losses can be read from Fig. 3.5 derived from Kays and London (1984) and Phillips (1990).

Figure 3.5(a) is applicable to the low aspect ratio channels,  $\alpha_c < 0.1$  and

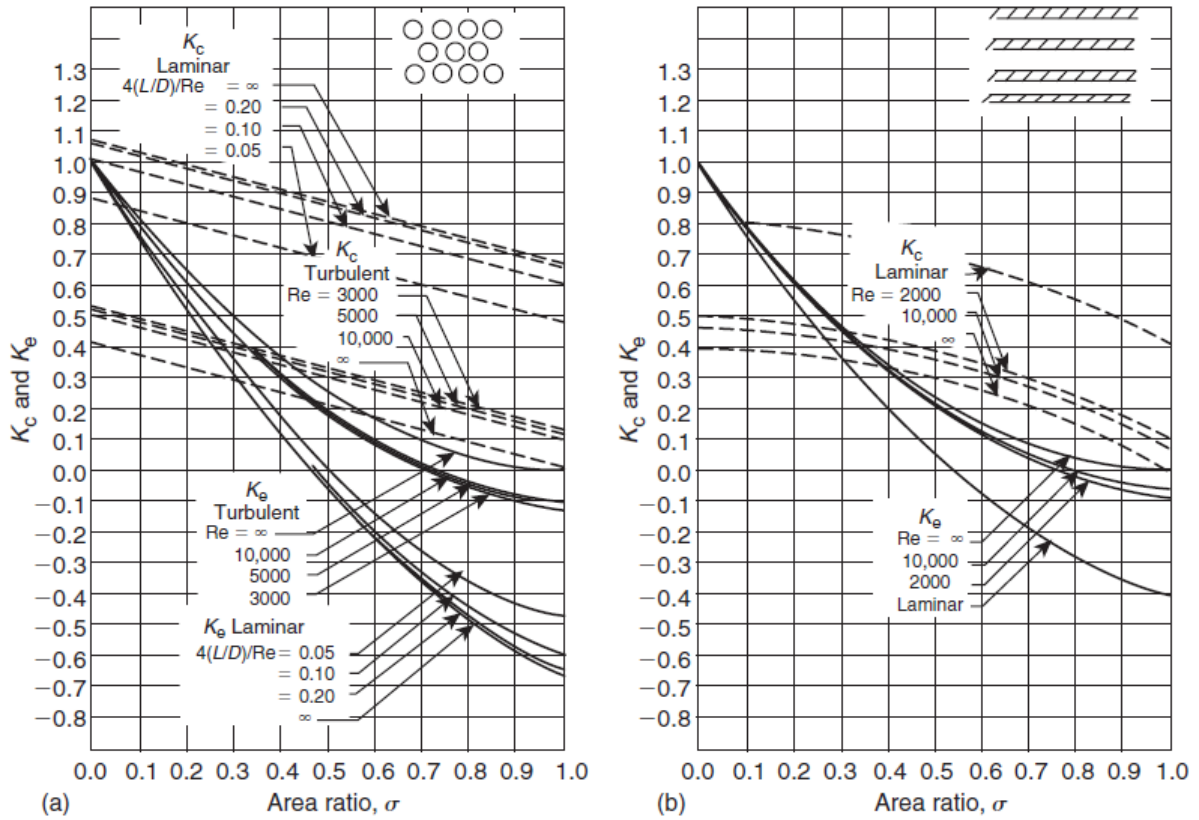
Fig. 3.5(b) is for  $0.1 \leq \alpha_c \leq 1.0$ .

Equation (3.24) can also be written in terms of the

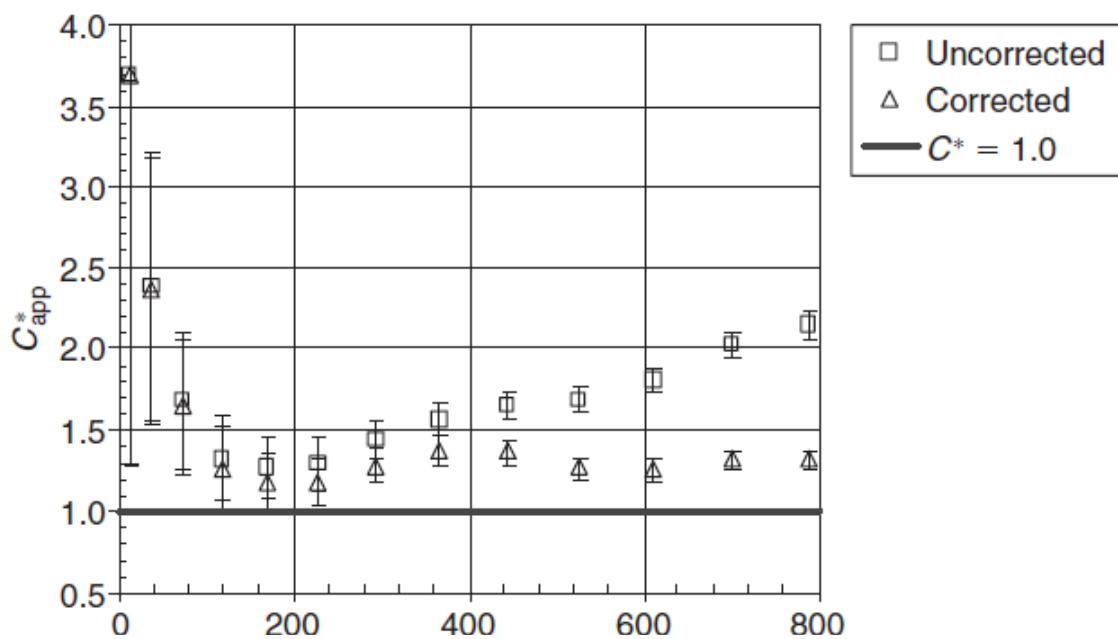
fully developed friction factor  $f$  and the pressure drop defect  $K(x)$ :

$$\Delta p = \frac{\rho u_m^2}{2} \left[ (A_c/A_p)^2 (2K_{90}) + (K_c + K_e) + \frac{4fL}{D_h} + K(x) \right] \quad (3.25)$$

For  $L > L_h$ ,  $K(x)$  is replaced by the Hagenbach's factor  $K(\infty)$  as discussed earlier in Section 3.2.3.



**Fig. 3.5.** Contraction and expansion loss coefficients for flow between inlet and outlet manifolds and the microchannels, Kays and London (1984).



**Fig. 3.6.** Variation of  $C_{app}^* = f_{app,ex} / f_{app,th}$  for a  $200 \mu\text{m}$  square microchannel array after correcting for entrance and exit losses.

Steinke and Kandlikar (2005a) conducted experiments on square silicon microchannels with 200  $\mu\text{m}$  sides and 10 mm in length, and analyzed the data in detail. A parameter  $C^*$  is introduced to represent the ratio of the

experimental and theoretical apparent friction factors.

Figure 3.6 shows the data before and after applying the corrections in

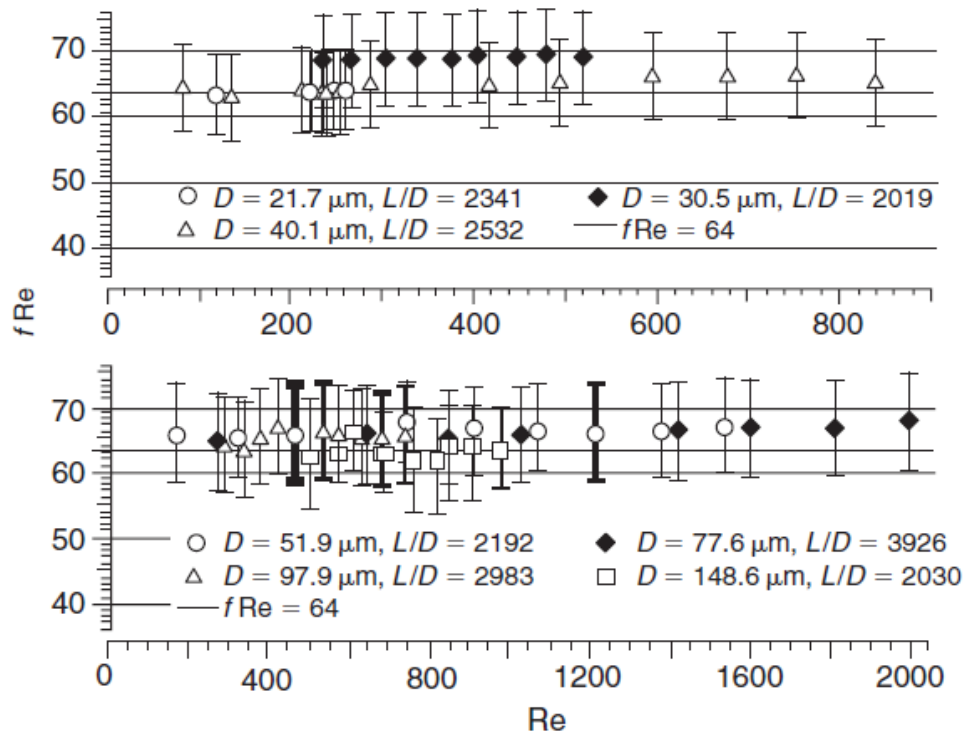
$$C^* = f_{\text{app,ex}} / f_{\text{app,th}},$$

where the subscripts ex and th refer to experimental and theoretical values respectively. The uncorrected data converges to within 30% of the theoretical predictions after all corrections are applied. It is suspected that the high errors are due to slight variations in the channel cross-sectional area over its length, different entrance conditions, and errors associated with interpolations during data reduction.

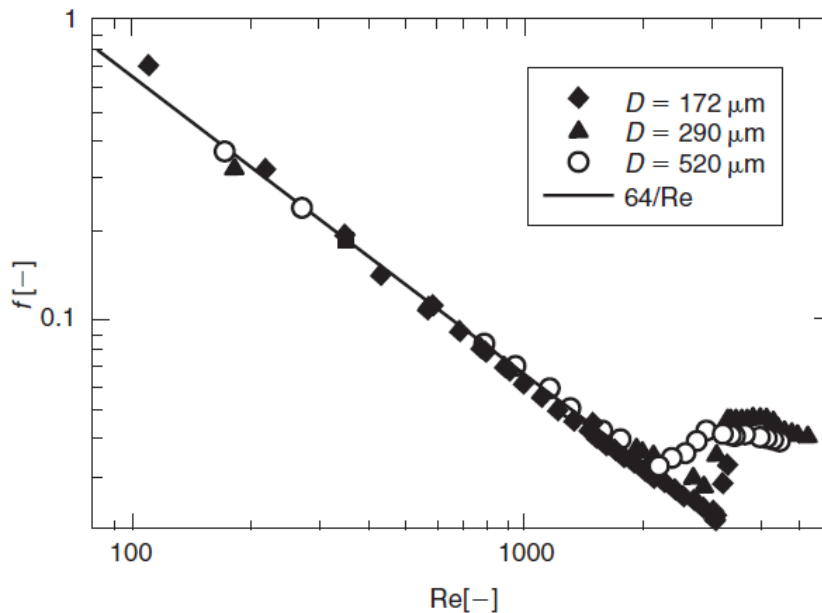
Figure 3.7 shows the excellent agreement between the laminar flow theory and experimental results obtained by Judy et al. (2002) in 15–150  $\mu\text{m}$  round and square microchannels made of fused silica and stainless steel with distilled water, methanol and isopropanol.

Similar agreement is seen in Fig. 3.8 by Bucci et al. (2004) in 172, 290 and 520  $\mu\text{m}$  diameter stainless steel circular tubes with water in the fully developed laminar flow region.

Niklas and Favre-Marinet (2003) analyzed the flow of water in a network of triangular microchannels with  $D_h = 110 \mu\text{m}$ . The contributions due to various losses were carefully analyzed in both their numerical as well as experimental work. They concluded that the classical theory is applicable to modeling the flow through the entire system.



**Fig. 3.7.** Comparison between theory and experimental data of Judy et al. (2002) in 50–150  $\mu\text{m}$  circular and square microchannels in the fully developed laminar region. (Note:  $f$  in this figure represents the **Darcy friction factor**, which is **four times** the **Fanning friction factor** used elsewhere in this text.)



**Fig. 3.8.** Comparison between theory and experimental data of Bucci et al. (2004) in 171–520  $\mu\text{m}$  diameter circular tubes in the fully developed laminar region. (Note:  $f$  in this figure represents the **Darcy friction factor**, which is **four times** the **Fanning friction factor** used elsewhere in this text.)

Another point that needs to be emphasized is the **large errors** that are associated with **microscale experiments** as pointed out by Judy et al. (2002). A detailed analysis was conducted by Steinke and Kandlikar (2005a) for rectangular microchannels using actual measured parameters. Their final expression for **uncertainty** in estimating the product  $fRe$  for **rectangular** channels is given by:

$$\frac{U_{fRe}}{fRe} = \left[ 2\left(\frac{U_\rho}{\rho}\right)^2 + \left(\frac{U_\mu}{\mu}\right)^2 + \left(\frac{U_{\Delta p}}{\Delta p}\right)^2 + \left(\frac{U_L}{L}\right)^2 + 3\left(\frac{U_Q}{Q}\right)^2 + 5\left(\frac{U_a}{a}\right)^2 + 5\left(\frac{U_b}{b}\right)^2 + 2\left(\frac{U_a}{a+b}\right)^2 + 2\left(\frac{U_b}{a+b}\right)^2 \right]^{1/2} \quad (3.26)$$

The details of the derivation are given by Steinke (2005).

(U: Uncertainty For instant  $U_\rho = \pm\Delta\rho$  – LK)

Note that the uncertainties in the **height  $a$**  and **width  $b$**  measurements have a

**major** influence on the **overall uncertainty**,

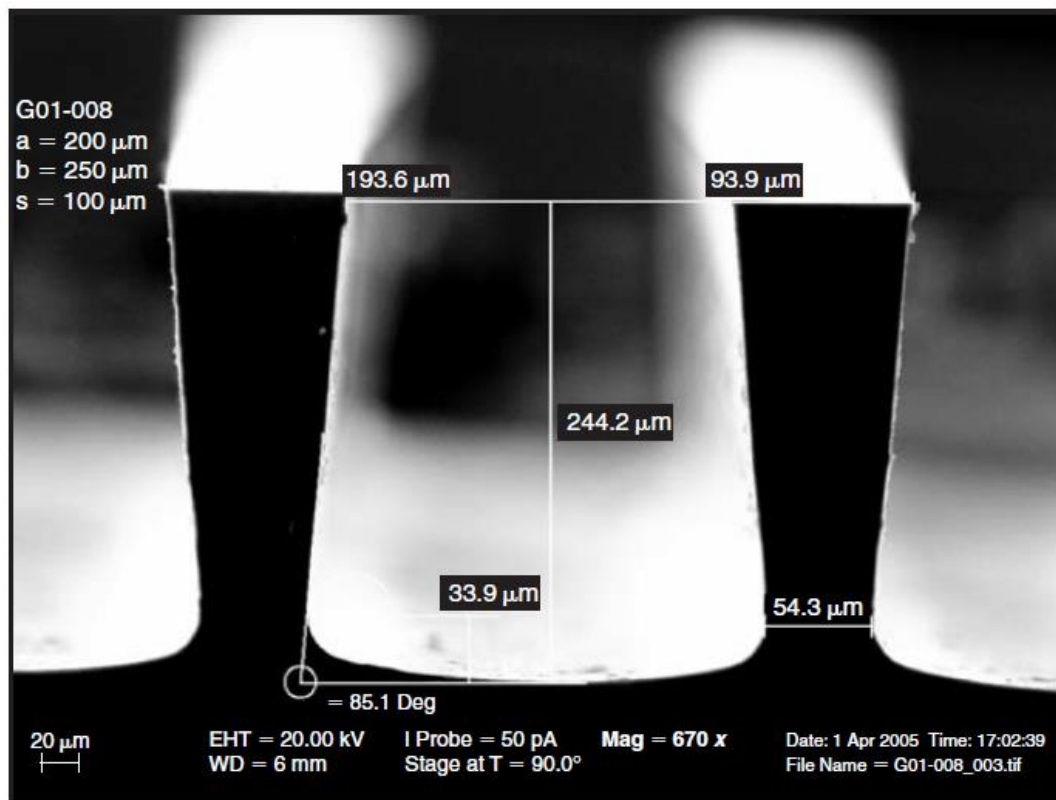
followed by the uncertainties in the

**density** and **volumetric flow rate** measurements.

The **low flow Reynolds number** data in **Fig. 3.6** exhibits **extremely large uncertainties** due to the errors associated with the **flow rate measurement**. The large discrepancy reported in literature by some of the earlier investigators is the result of the uncertainties in channel dimensions and flow rate measurements, entrance and exit losses, and the developing region effects.

**Figure 3.9** shows a picture of a microchannel cross-section obtained with a scanning electron microscope by Steinke and Kandlikar (2005a). Note that the actual channel profile deviates significantly from the intended rectangular profile. The side walls are undercut by about 20  $\mu\text{m}$  and the corners are rounded. This illustrates the need to accurately measure the flow area and the for estimating the pressure drop characteristics of microchannels.

Similar conclusions were reached by other investigators, for example, Li et al. (2000), Celata (2002), Judy et al. (2002), Baviere et al. (2004), and Tu and Hrnjak (2003). The increasing deviations from the theoretical values at higher Reynolds numbers in many data sets reported in literature is believed to be due to the increased length of the developing region. Baviere and Ayela (2004) measured the local pressures along the flow in a microchannel of height  $7.5\text{ }\mu\text{m}$ . The range of Reynolds numbers considered was between 0.1 and 15. Their results were found to agree with the classical theory. They also identified the errors associated with the channel height measurement as being the largest source of uncertainty.



**Fig. 3.9.** Cross-sectional view of a rectangular microchannels etched in silicon showing distortion in channel geometry from ideal rectangular profile, Steinke and Kandlikar (2005a).

Baviere et al. (2004) conducted experiments with bronze, altuglas, and silicon coated microchannels with channel heights between 7.1 and 300  $\mu\text{m}$  and channel widths between 1 and 25 mm. Their results were also in excellent agreement with the classical theory.

Other effects that may influence the flow in microchannels are

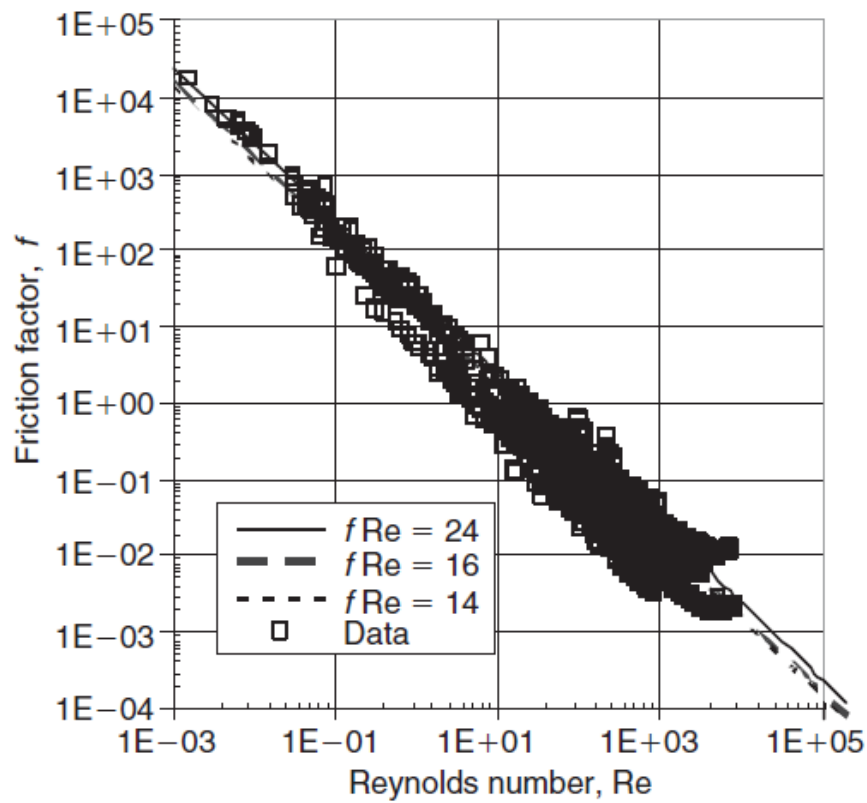
(i) high viscous dissipation causing a change in the fluid viscosity at the wall, and

(ii) electrokinetic effects,

which are covered in Chapter 4.

The effect of viscous dissipation is reviewed by Shen et al. (2004). Xu et al. (2000) modeled this effect and found that the velocity profile was modified due to the viscous dissipation. The resulting friction factors were predicted to be lower due to a reduction in the viscosity at higher liquid temperatures. The viscous effect was studied experimentally by Judy et al. (2002). Their pressure drop data correlated well with the theory when they used the average of the inlet and outlet fluid temperatures from their stainless steel microtubes, which were 15–100  $\mu\text{m}$  in diameter. Koo and Kleinstreuer (2004) concluded that viscous dissipation effects increase rapidly with a decrease in the channel dimensions and hence should be considered along with the imposed heat sources at the channel walls.

Figure 3.10 shows a plot of the friction factor versus Reynolds number using the experimental data points reported in literature as compiled by Steinke and Kandlikar (2005a). The lines represent the theoretical values for different channel aspect ratios. These data points were screened to eliminate the data that were obtained with large uncertainties and those which did not account for the entrance and exit losses. It is seen that the general trends are followed, though the errors are still quite large mainly due to the uncertainties associated with channel size measurement.



**Fig. 3.10.** Comparison of friction factors reported in literature with the theoretical values for different microchannel geometries. Entrance and exit losses and the developing region at the entrance were not accounted for in a number of data sets reviewed by Steinke and Kandlikar (2005a).

### 3.3.2. Laminar-to-turbulent transition

The laminar-to-turbulent flow transition is another topic that was analyzed by a number of investigators.

The laminar-to-turbulent transition in abrupt entrance rectangular ducts was found to occur at a

transition Reynolds number of  $Re_t = 2200$  for  $\alpha_c = 1$

and at

$Re_t = 2500$  (Hartnett et al., 1962) for parallel plates with  $\alpha_c = 0$ .

For other aspect ratios, a linear interpolation between these two values is recommended.

Some of the initial studies indicated an early transition to turbulent flow in microchannels. However, a number of recent studies showed that the laminar-to-turbulent transition remains unchanged. Figure 3.8 by Bucci et al. (2004) shows that the transition occurred around  $Re_t = 2000$  for circular microtubes 171–520  $\mu\text{m}$  in diameter. The results of Baviere et al. (2004) also indicate that the laminar-to-turbulent transition in smooth microchannels is not influenced by the channel dimensions and occurs around 2300. Similar results were reported by a number of investigators, including Bucci et al. (2004), Schmitt and Kandlikar (2005), Kandlikar et al. (2005) for minichannels with  $D_h < 1 \text{ mm}$ ; and Li et al. (2000) for  $80 \mu\text{m} \leq D_h \leq 166.3 \mu\text{m}$ .

The transition from the laminar-to-turbulent region is influenced by the channel surface roughness. Further details of this effect are discussed in the next section on roughness.

## 3.4. Roughness effects

### 3.4.1. Roughness representation

Darcy (1857) investigated the effects of surface roughness on the turbulent flow of water in rough pipes made of cast iron, lead, wrought iron, and asphalt-covered cast iron. The pipes were 12 to 500 mm in diameter and 100 m long.

Fanning (1886) proposed a correlation for the pressure drop as a function of roughness.

Mises (1914) is credited with introducing the term **relative roughness**, which was originally defined as the **ratio** of **absolute roughness** to the **pipe radius** (we now use pipe diameter to normalize roughness parameters, such as average roughness  $R_a$ ).

Nikuradse (1933) presented a comprehensive review of the literature covering uniform roughness and roughness structures such as corrugations, and some of the available correlations relating friction factor to relative roughness. He also conducted a systematic study on friction factor by applying uniform diameter sand grain particles with Japanese lacquer to the inner pipe surface and measuring the pressure drop. He **identified three regions** which form the **basis of** the current **Moody diagram** that is used for friction factor estimation.

The term **relative roughness** is used for the ratio  $\varepsilon/D_h$  where  $\varepsilon$  is the **average roughness**.

Nikuradse used the diameter of the uniform sand grain particles to represent the roughness  $\varepsilon$ . The range of relative roughness in Nikuradse's experiments on circular pipes was  $0.001 \leq \varepsilon/D \leq 0.033$ .

Furthermore, Nikuradse identified three ranges to describe his laws of resistances.

In Range I, for low Reynolds numbers corresponding to laminar flow, the friction factor was independent of surface roughness and was given by the now established classical equation  $f = 16/\text{Re}$  for circular pipes.

In Range II, he described the roughness effects on the friction factor in turbulent (high Reynolds numbers-LK) flow by

$$f = 0.079/\text{Re}^{1/4},$$

and finally in Range III, the friction factor is independent of the Reynolds number but depends on the relative roughness,

$$f = 0.25/[1.74 + 2 \log(2\varepsilon/D)]^2.$$

Note that the original equations by Nikuradse have been modified to express them in terms of the Fanning friction factor (which is one fourth of Darcy friction factor,

$$f = 0.25 f_{\text{Darcy}}).$$

Moody (1944) used the available data on the friction factor and developed the well-known Moody diagram covering the relative roughness range of

$$0 \leq \varepsilon/D \leq 0.05.$$

Following the earlier investigators, he used the tube root diameter of a circular pipe, similar to Nikuradse's work, in expressing the relative roughness  $\varepsilon/D$ .

A number of different parameters were studied for their suitability in representing the roughness effects. Kandlikar et al. (2005) investigated parameters based on various roughness characterization schemes and proposed a set of three parameters:

Mean Profile Peak Height ( $R_p$ ),

Mean Spacing of Profile Irregularities ( $RS_m$ ), and the

Floor Distance to Mean Line ( $F_p$ ),

as shown in Fig. 3.11. Two of these parameters ( $R_p$  and  $RS_m$ ) are defined in the ASME B46.1-2002 and the other ( $F_p$ ) is proposed by Kandlikar et al. (2005).

- **Average maximum profile peak height ( $R_{pm}$ )**: The distance between the average of the individual highest points of the profile ( $R_{p,i}$ ) and the mean line within the evaluation length. The mean line represents the conventional average roughness value ( $R_a$ ).
- **Mean spacing of profile irregularities ( $RS_m$ )**: Consists of the mean value of the spacing between profile irregularities within the evaluation length. The irregularities of interest are the peaks, so this is equivalent to the Pitch.

$$RS_m = \frac{1}{n} \sum_{i=1}^n S_{m_i} \quad (3.27)$$

- **Floor distance to mean line ( $F_p$ )**: Consists of the distance between the main profile mean line (determined by  $R_a$ ) and the floor profile mean line. The floor profile is the portion of the main profile that lies below the main profile mean line.

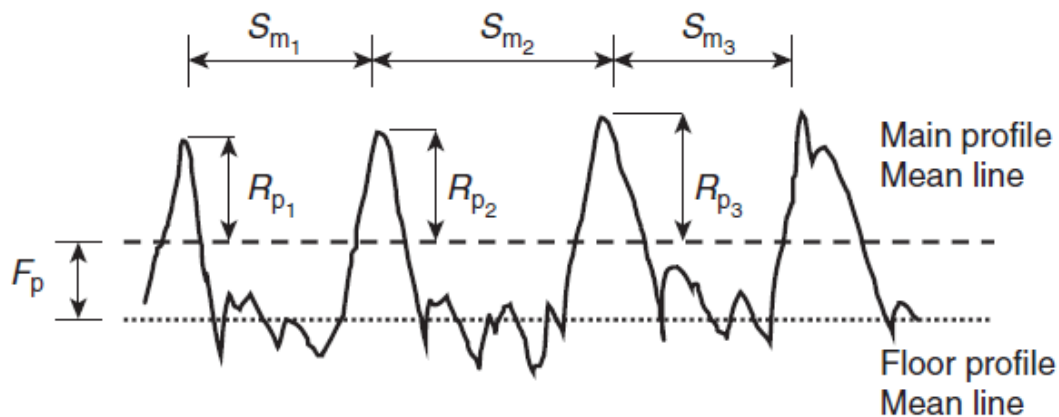


Fig. 3.11. Maximum profile peak height ( $R_p$ ), Mean spacing of profile irregularities ( $RS_m$ ), and Floor distance to mean line ( $F_p$ ).

The three parameters described above allow the characterization of the peak height, peak spacing and the distance from the floor to the mean line.

These parameters will define the characteristics of the surface roughness that influence the location and shape of the fluid flow streamlines (as described by Webb et al., 1971 and Kandlikar et al., 2005) and, consequently, the size of the recirculation flow zones between roughness elements.

From the above parameters, the equivalent roughness  $\varepsilon$  can be estimated by the following relationship:

$$\varepsilon = R_{pm} + F_p \quad (3.28)$$

For the sand grain roughness employed by Nikuradse, the new definition of roughness  $\varepsilon$  yields the same value, and hence no correction is needed for any of the friction factor correlations and charts. The use of mean spacing between profile irregularities is expected to represent the structured roughness surfaces. Such surfaces may be designed in the future to obtain specific pressure drop and heat transfer performance characteristics.

### 3.4.2. Roughness effect on friction factor

For microchannels, the relative roughness values are expected to be higher than the limit of 0.05 used in the Moody diagram.

Kandlikar et al. (2005) considered the effect of cross-sectional area reduction due to protruding roughness elements and recommended using the constricted flow area in calculating the friction factor.

Using a constricted diameter  $D_{cf} = D - 2\varepsilon$ , a modified Moody diagram was presented as shown in Fig. 3.12.

In the turbulent region, it was found that such a representation yielded a constant value of friction factor above  $\varepsilon/D_{cf} > 0.03$ .

In the turbulent fully rough region,  $0.03 \leq \varepsilon/D_{cf} \leq 0.05$ , the friction factor based on the constricted flow diameter is given by:

$$f_{\text{Darcy,cf}} = 0.042 \quad (3.29a)$$

In terms of the Fanning friction factor, we get:

$$f_{cf} = f_{\text{Darcy,cf}}/4 = 0.042/4 = 0.0105 \quad (3.29b)$$

Since experimental data is not available beyond  $\varepsilon/D_{cf} > 0.05$ , using Eq. (3.29) for higher relative roughness values than 0.05 is not recommended.

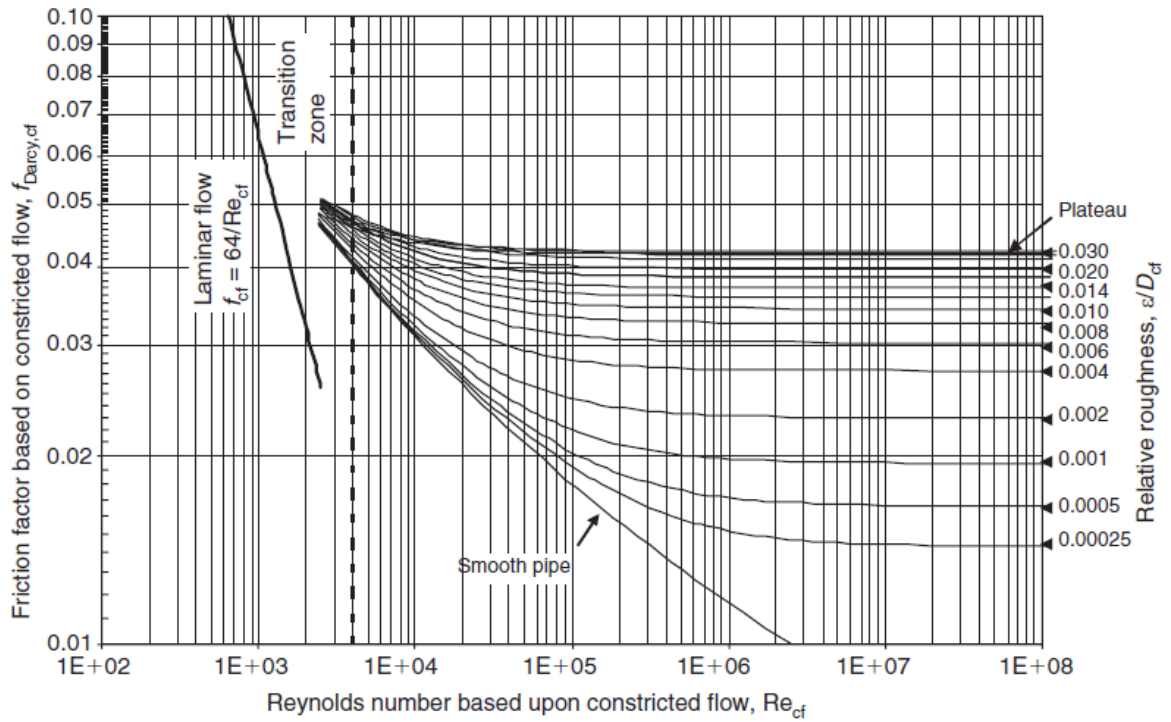
Note that the friction factor and the geometrical and flow parameters are based on the constricted flow diameter as given by the following equations:

$$D_{cf} = D - 2\varepsilon \quad (3.30)$$

$$\Delta p = \frac{2f_{cf}\rho u_{m,cf}^2 L}{D_{h,cf}} \quad (3.31)$$

$$u_{m,cf} = \dot{m}/A_{cf} \quad (3.32)$$

$$\text{Re}_{cf} = \frac{\rho u_{m,cf} D_{h,cf}}{\mu} \quad (3.33)$$



**Fig. 3.12.** Darcy friction factor plot based on a constricted flow diameter, Kandlikar et al. (2005).

In the fully developed laminar flow region, the constricted friction factor is given by the following equation, Kandlikar et al. (2005):  
Laminar region,  $0 \leq \varepsilon/D_{h,cf} \leq 0.15$ ;

$$f_{cf} = \frac{Po}{Re_{cf}} \quad (3.34)$$

where the Poiseuille number  $Po$  is given by either Eq. (3.9) or (3.10) depending on the channel geometry.

$$Po = fRe = 16 \quad (3.9)$$

$$\alpha_c = a/b$$

$$Po = fRe = 24(1 - 1.3553\alpha_c + 1.9467\alpha_c^2 - 1.7012\alpha_c^3 + 0.9564\alpha_c^4 - 0.2537\alpha_c^5) \quad (3.10)$$

In the turbulent region, use the modified Miller equation as recommended by Kandlikar et al. (2005):

Fully developed turbulent region,  $0 \leq \varepsilon/D_{h,cf} < 0.03$ ;

$$f_{cf} = \frac{0.25 \left[ \log_{10} \left( \frac{\varepsilon/D_{h,cf}}{3.7} + \frac{5.74}{Re_{cf}^{0.9}} \right) \right]^{-2}}{4} \quad (3.35)$$

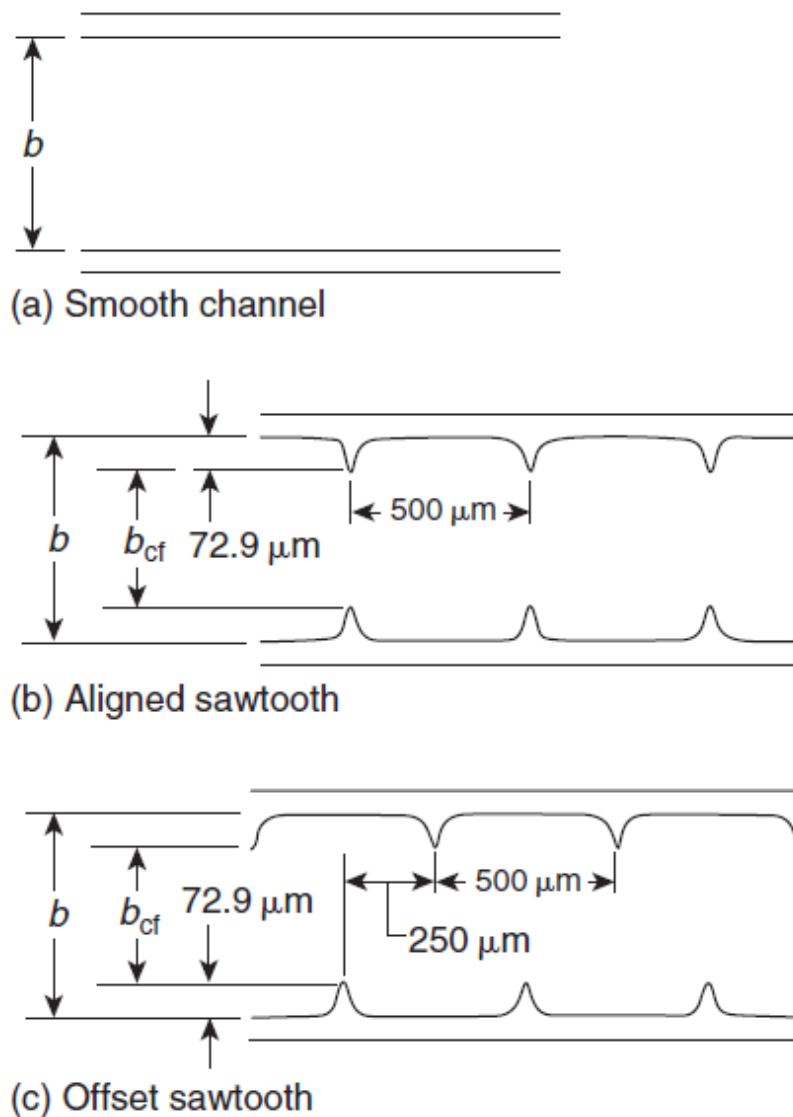
Fully developed turbulent region,  $0.03 \leq \varepsilon/D_{h,cf} \leq 0.05$

$$f_{cf} = 0.0105 \quad (3.36)$$

No recommendation can be made for the turbulent region with  $\varepsilon/D_{h,cf} > 0.05$  because of the lack of detailed experimental data in literature for this region. Further experimental work in this area is strongly recommended.

For rectangular channels with all sides having a roughness  $\varepsilon$ , the mean flow velocity is calculated using the constricted flow area and the constricted channel dimensions of  $a_{cf} = a - 2\varepsilon$  and  $b_{cf} = b - 2\varepsilon$ . The hydraulic diameter is calculated using the constricted channel dimensions.

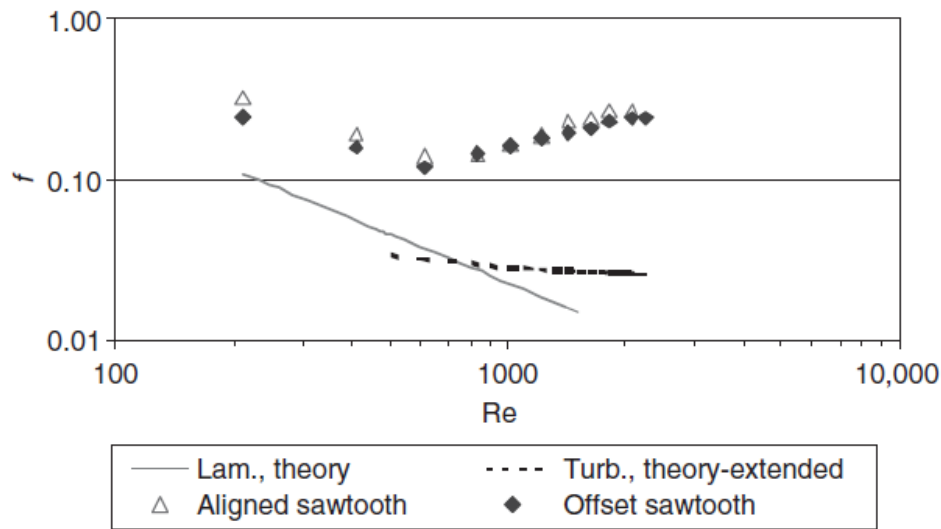
Schmitt and Kandlikar (2005) conducted experiments to study the effect of a large relative roughness of  $\varepsilon/D_{h,cf}$  up to 0.15 in smooth and artificially roughened rectangular minichannels. They introduced sawtooth roughness elements in a 10.3 mm wide and 100 mm long test channel. Cross-sectional views of the smooth channels and channels with sawtooth roughness elements are shown in Fig. 3.13. The channel gap  $b_{cf}$  was varied to produce values of  $\varepsilon/D_{h,cf}$  from 0.03 to 0.15. Differential pressure taps for measuring the local pressure drops were located at several locations along the flow length in the fully developed region.



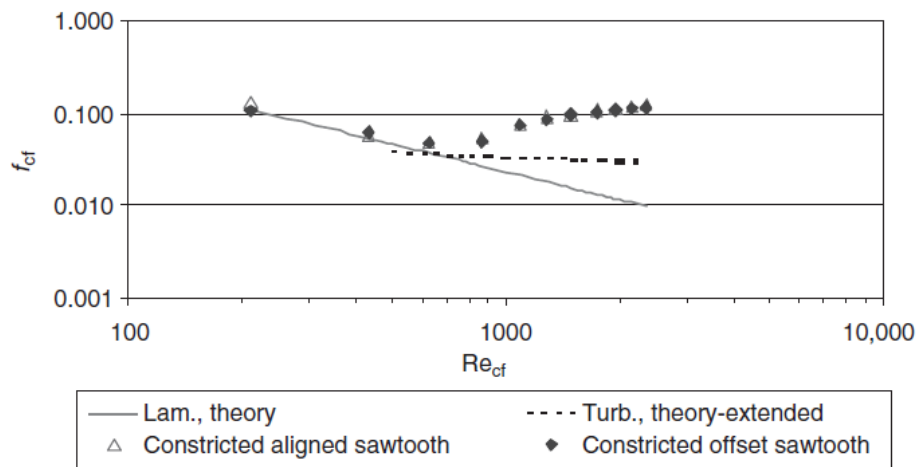
**Fig. 3.13.** Roughness elements used by Schmitt and Kandlikar (2005) and Kandlikar et al. (2005).

The smooth channel results reported by Kandlikar et al. (2005) closely follow the classical laminar flow theory for friction factor. The experimental data from the two sawtooth structures are represented in Fig. 3.14 for flow of water with  $b = 500 \mu\text{m}$ , resulting in a  $D_h$  of  $953 \mu\text{m}$ . This plot uses the height  $b$  in calculating  $D_h$  and  $\text{Re}$ . The dashed line represents the  $f \text{Re} = \text{Constant}$  line corresponding to the channel aspect ratio  $\alpha_c = a/b$ . It is seen that the agreement in the low Reynolds number region corresponding to laminar flow is considerably off.

Figure 3.15 shows the same data used in Fig. 3.14, but plotted with the constricted flow areas and using the constricted flow parameters given by Eqs. (3.30)–(3.33). The results shown in Fig. 3.15 in the laminar region show good agreement with the accepted laminar flow theory ( $f_{cf} \text{Re}_{cf} = \text{Constant}$ ). Similar results were obtained for different gap sizes (yielding different relative roughness values) for both the offset sawtooth and aligned sawtooth geometries.



**Fig. 3.14.** Fully developed friction factor versus Reynolds number, both based on hydraulic diameter for water flow.  $D_h = 953 \mu\text{m}$ ,  $b = 500 \mu\text{m}$ ,  $b_{cf} = 354 \mu\text{m}$ ,  $w = 10.03 \text{ mm}$ ,  $\varepsilon/D_h = 0.0735$ , Kandlikar et al. (2005).



**Fig. 3.15.** Fully developed friction factor versus Reynolds number, both based on constricted flow hydraulic diameter; water flow.  $D_{h,cf} = 684 \mu\text{m}$ ,  $b = 500 \mu\text{m}$ ,  $b_{cf} = 354 \mu\text{m}$ ,  $w = 10.03 \text{ mm}$ ,  $\varepsilon/D_{h,cf} = 0.1108$ , Kandlikar et al. (2005).

### 3.4.3. Roughness effect on the laminar-to-turbulent flow transition

The transition from laminar-to-turbulent flow has been reported to occur in microchannels at Reynolds numbers considerably below 2300.

In many experiments, the transition has been mistakenly identified to occur early based on experimental data uncorrected for the developing length (Steinke and Kandlikar, 2005a). Kandlikar et al. (2003) conducted experiments with stainless steel tubes and noted that early transition occurred for a 0.62 mm ID stainless steel tube with a surface roughness  $\varepsilon/D_h$  of 0.355%. Schmitt and Kandlikar (2005) conducted careful experiments with plain and sawtooth roughened channels with air and water. Their results for smooth rectangular channels showed a transition Reynolds number between 2000 and 2300,

but for

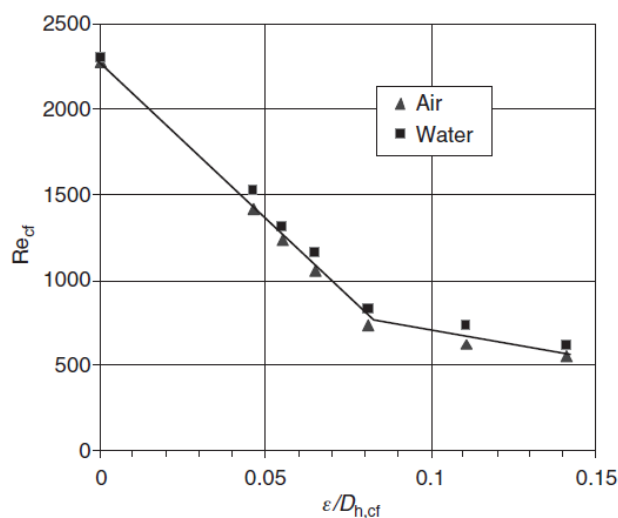
increasing relative roughness values there was a decreasing transition Reynolds numbers

as seen in Figs. 3.14 and 3.15. Figure 3.16 shows the transition Reynolds number as a function of the relative roughness (based on the constricted hydraulic diameter). The following equations are used to describe the roughness effects based on their experimental data.

Laminar-to-turbulent transition criteria:

$$\text{For } 0 < \varepsilon/D_{h,cf} \leq 0.08: \quad Re_{t,cf} = 2300 - 18,750(\varepsilon/D_{h,cf}) \quad (3.37)$$

$$\text{For } 0.08 < \varepsilon/D_{h,cf} \leq 0.15: \quad Re_{t,cf} = 800 - 3270(\varepsilon/D_{h,cf} - 0.08) \quad (3.38)$$



**Fig. 3.16.** Transition Reynolds number variation with relative roughness based on constricted flow diameter, Equations (3.37) and (3.38) plotted along with data from Schmitt and Kandlikar (2005).

Friction factor in the transition region for rough tubes with  $\varepsilon/D_{h,cf} \leq 0.05$  at a given  $Re_{cf}$  can be obtained by a linear interpolation of (i) the laminar friction factor obtained from Eq. (3.34) at the transition Reynolds number, and (ii) the turbulent friction factor at  $Re_{cf} = 2300$  given by Eq. (3.35) or (3.36) depending on the value of  $\varepsilon/D_{h,cf}$ . For  $\varepsilon/D_{h,cf} > 0.05$  in the transition region, additional experimental data is needed before any recommendations can be made.

#### **3.4.4. Developing flow in rough tubes**

Developing flow in rough tubes has not been explored in literature.

As a preliminary estimate, the methods described in Section 3.2.3 for smooth tubes are recommended by introducing the constricted flow diameter in calculating the flow velocity, Reynolds number and friction factor. Further research in this area is needed.

## 3.5. Heat transfer in microchannels

### 3.5.1. Fully developed laminar flow

The Nusselt number in fully developed laminar flow is expected to be constant as predicted by the classical theory.

However, there are a number of investigations reported in literature that show a trend (Heat Transfer-LK) increasing with Reynolds number in this range. This results from the experimental uncertainties as discussed in Section 3.5.3.

The Nusselt number in the fully developed laminar flow is constant and depends on the channel geometry and the wall heat transfer boundary condition.

Table 3.1 presents the Nusselt numbers for commonly used geometries under constant heat flux and constant wall temperature boundary conditions.

For a rectangular channel, the Nusselt number depends on the channel aspect ratio  $\alpha_c = a/b$ , and the wall boundary conditions. Three boundary conditions are identified in literature and the Nusselt number for each one is given below.

Constant wall temperature, T-boundary condition:

$$Nu_T = 7.541(1 - 2.610\alpha_c + 4.970\alpha_c^2 - 5.119\alpha_c^3 + 2.702\alpha_c^4 - 0.548\alpha_c^5) \quad (3.39)$$

Constant circumferential wall temperature, uniform axial heat flux, H1 boundary condition:

$$Nu_{H1} = 8.235(1 - 2.0421\alpha_c + 3.0853\alpha_c^2 - 2.4765\alpha_c^3 + 1.0578\alpha_c^4 - 0.1861\alpha_c^5) \quad (3.40)$$

Constant wall heat flux, both circumferentially and axially:

$$Nu_{H2} = 8.235(1 - 10.6044\alpha_c + 61.1755\alpha_c^2 - 155.1803\alpha_c^3 + 176.9203\alpha_c^4 - 72.9236\alpha_c^5) \quad (3.41)$$

In reality, all practical situations fall somewhere in the middle of these three boundary conditions. This becomes an especially important issue in the case of microchannels because of the difficulty in identifying a correct boundary condition with discretely spaced heat sources, and two-dimensional effects in the base and the fins.

The heating in microchannel geometries generally comes from three sides, as a cover of glass or some other material is bonded on top of the microchannels to form the flow passages. The fully developed Nusselt numbers for both three- and four-side heated ducts have been compiled from various sources, including Wibulswas (1966), and Phillips (1987) and are given in Table 3.3. Note that the side with dimension  $a$  is not heated, and the channel aspect ratio is defined as  $\alpha_c = a/b$ .

**Table 3.3**  
Fully developed laminar flow Nusselt numbers.

| $\alpha_c = a/b$ | $Nu_{fd,3}$ | $Nu_{fd,4}$ |
|------------------|-------------|-------------|
| 0                | 8.235       | 8.235       |
| 0.10             | 6.939       | 6.700       |
| 0.20             | 6.072       | 5.704       |
| 0.30             | 5.393       | 4.969       |
| 0.40             | 4.885       | 4.457       |
| 0.50             | 4.505       | 4.111       |
| 0.70             | 3.991       | 3.740       |
| 1.00             | 3.556       | 3.599       |
| 1.43             | 3.195       | 3.740       |
| 2.00             | 3.146       | 4.111       |
| 2.50             | 3.169       | 4.457       |
| 3.33             | 3.306       | 4.969       |
| 5.00             | 3.636       | 5.704       |
| 10.00            | 4.252       | 6.700       |
| > 10.00          | 5.385       | 8.235       |

$a$  – unheated side in three-side heated case.

For intermediate values, use the curve-fit equations provided in Appendix A at the end of the chapter.

### 3.5.2. Thermally developing flow

The thermal entry length is expressed by the following form for flow in ducts:

$$\frac{L_t}{D_h} = c \text{ Re Pr} \quad (3.42)$$

For circular channels, the leading constant  $c$  in Eq. (3.42) is found to be

$$c = 0.05,$$

while for rectangular channels, the plots presented by Phillips (1987) suggest

$$c = 0.1.$$

The local heat transfer in the developing region of a circular tube is given by the following equations (Shah and London, 1978):

$$\text{Nu}_x = 4.363 + 8.68(10^3 x^*)^{-0.506} e^{-41x^*} \quad (3.43)$$

$$x^* = \frac{x/D_h}{\text{Re Pr}} \quad (3.44)$$

For rectangular channels with the four-side heating configuration, Nusselt numbers in the thermally developing region presented in Table 3.4 are derived from Phillips' (1987) work.

For the three-side heating configuration the following scheme is suggested by Phillips (1990).

Three-side heating,  $\alpha_c \leq 0.1$  and  $\alpha_c \geq 10$ , use four-side heating table without any modification.

Three-side heating,  $0.1 < \alpha_c < 10$ :

$$\text{Nu}_{x,3}(x^*, \alpha_c) = \text{Nu}_{x,4}(x^*, \alpha_c) \frac{\text{Nu}_{fd,3}(x^* = x_{fd}^*, \alpha_c)}{\text{Nu}_{fd,4}(x^* = x_{fd}^*, \alpha_c)} \quad (3.45)$$

The subscripts  $x,3$  and  $x,4$  refer to the location at a distance  $x$  in the heated length for the three-sided and four-sided heating cases respectively.

The Nusselt numbers in the fully developed region for both heating configurations are obtained from Table 3.3, and

in the developing region Nusselt numbers for the four-sided heating are obtained from Table 3.4.

The use of fully developed hydrodynamic conditions in heat transfer analysis is reasonable for water. Garimella and Singhal (2004) noted that assuming fully developed hydrodynamic conditions and thermally developing conditions resulted in a satisfactory agreement with their data for microchannels.

**Table 3.4**  
Thermal entry region Nusselt numbers.

| $x^*$   | $\alpha_c \leq 0.1^*$ | $Nu_{x,4}$        |                    |                  |                  |                         |
|---------|-----------------------|-------------------|--------------------|------------------|------------------|-------------------------|
|         |                       | $\alpha_c = 0.25$ | $\alpha_c = 0.333$ | $\alpha_c = 0.5$ | $\alpha_c = 1.0$ | $\alpha_c \geq 10^{**}$ |
| 0.0001  | 31.4                  | 26.7              | 27.0               | 23.7             | 25.2             | 31.6                    |
| 0.0025  | 11.9                  | 10.4              | 9.9                | 9.2              | 8.9              | 11.2                    |
| 0.005   | 10                    | 8.44              | 8.02               | 7.46             | 7.1              | 9.0                     |
| 0.00556 | 9.8                   | 8.18              | 7.76               | 7.23             | 6.86             | 8.8                     |
| 0.00625 | 9.5                   | 7.92              | 7.5                | 6.96             | 6.6              | 8.5                     |
| 0.00714 | 9.3                   | 7.63              | 7.22               | 6.68             | 6.32             | 8.2                     |
| 0.00833 | 9.1                   | 7.32              | 6.92               | 6.37             | 6.02             | 7.9                     |
| 0.01    | 8.8                   | 7                 | 6.57               | 6.05             | 5.69             | 7.49                    |
| 0.0125  | 8.6                   | 6.63              | 6.21               | 5.7              | 5.33             | 7.2                     |
| 0.0167  | 8.5                   | 6.26              | 5.82               | 5.28             | 4.91             | 6.7                     |
| 0.025   | 8.4                   | 5.87              | 5.39               | 4.84             | 4.45             | 6.2                     |
| 0.033   | 8.3                   | 5.77              | 5.17               | 4.61             | 4.18             | 5.9                     |
| 0.05    | 8.25                  | 5.62              | 5.00               | 4.38             | 3.91             | 5.55                    |
| 0.1     | 8.24                  | 5.45              | 4.85               | 4.22             | 3.71             | 5.4                     |
| 1       | 8.23                  | 5.35              | 4.77               | 4.11             | 3.6              | 5.38                    |

$x^* = x/(\text{Re Pr} D_h)$ ; \* – parallel plates, both sides heated; \*\* – parallel plates, one side heated.  
For intermediate values, use the curve-fit equations provided in Appendix A.

### 3.5.3. Agreement between theory and available experimental data on laminar flow heat transfer

Laminar flow heat transfer in microchannels has been studied by a number of researchers. Agreement with the classical laminar flow theory is expected to hold, but the reported data show significant scatter due to difficulties encountered in making accurate local heat flux and temperature measurements.

Steinke and Kandlikar (2005b) reviewed the available data and presented a comprehensive table showing the range of parameters employed.

The laminar flow heat transfer in the fully developed region is expected to be constant,

but the data taken from literature show a generally linear increase in Nusselt number with flow Reynolds number as seen in Fig. 3.17.

The main reasons for this discrepancy have been attributed to the following factors:

(i) **Entrance region effects:** The researchers have used the fully developed theoretical values for Nusselt numbers in their comparison with the experimental data. Due to the relatively short lengths employed in microchannels, the influence of the entrance region cannot be neglected.

The entrance region effects become more significant at higher Reynolds numbers, in part explaining the increasing Nusselt number trend with Reynolds number as seen in Fig. 3.17.

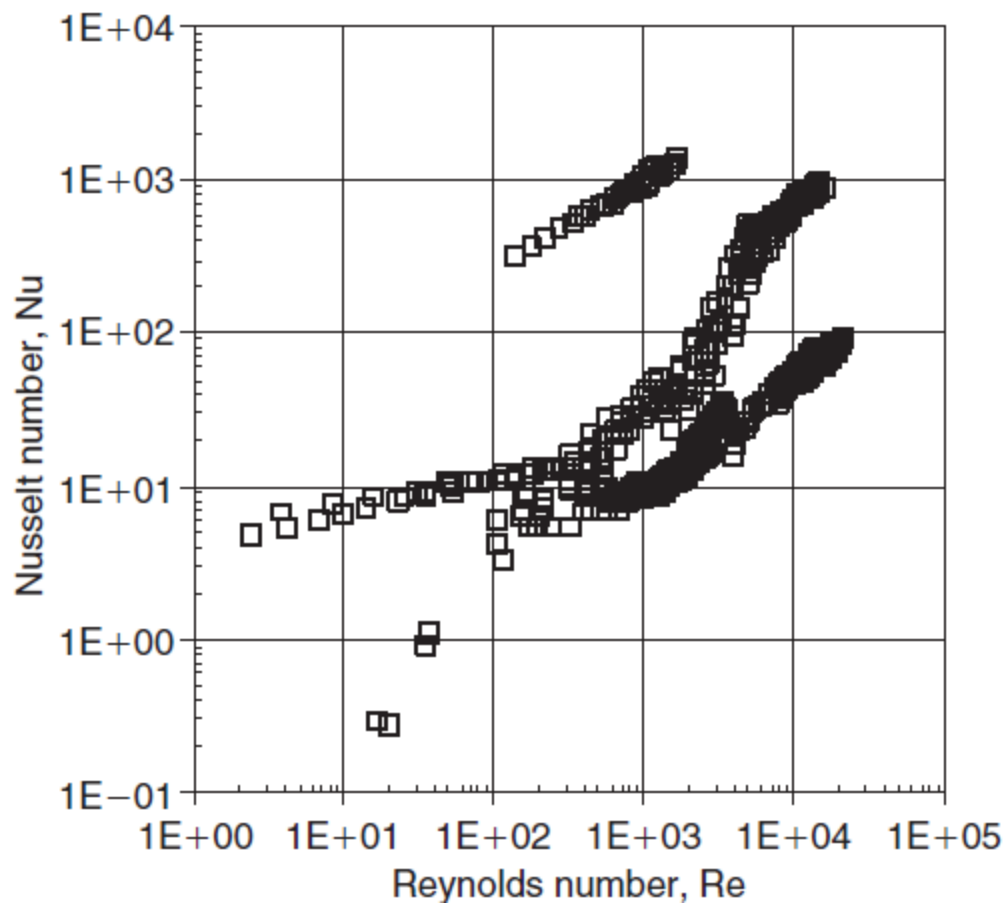
(ii) **Uncertainties in experimental measurements:** The uncertainties in heat transfer measurements have been analyzed by a number of investigators, including Judy et al. (2002) and Steinke and Kandlikar (2005b). The following equation for the uncertainty in the Nusselt number calculation from the uncertainties in the experimental measurements is presented by Steinke and Kandlikar for a rectangular channel of base width  $a$  and height  $b$ :

$$U_{Nu} = Nu \cdot \left[ \left( \frac{U_{k_f}}{k_f} \right)^2 + \left( \frac{U_I}{I} \right)^2 + \left( \frac{U_V}{V} \right)^2 + 4 \left( \frac{U_{T_s}}{T_s} \right)^2 + 2 \left( \frac{U_{T_i}}{T_i} \right)^2 + 2 \left( \frac{U_{T_o}}{T_o} \right)^2 + 3 \left( \frac{U_L}{L} \right)^2 + 4 \left( \frac{U_a}{a} \right)^2 + 5 \left( \frac{U_b}{b} \right)^2 + 2 \left( \frac{U_{\eta_f}}{\eta_f} \right)^2 \right]^{1/2} \quad (3.46)$$

where  $I$  – current,  $V$  – voltage,  $T_s$  – surface temperature,  $T_i$  – fluid inlet temperature,  $T_o$  – fluid outlet temperature,  $L$  – flow length,  $a$  and  $b$  – cross-section dimensions, and  $\eta_f$  – fin efficiency. The **uncertainties in the measurement** of the **surface temperature** and **flow channel dimensions** play a critical role in the **overall uncertainty**. The importance of accurate geometrical measurements was emphasized in the friction factor estimation as well. Accurate surface temperature measurement poses a significant challenge due to the small dimensions of the test section. Use of silicon chips with integrated circuits to measure the temperatures is recommended.

**Another factor** that makes the temperature measurements critical in the overall uncertainty is the **small temperature difference** between the surface and the fluid at the outlet. Since the heat transfer coefficients are very high in microchannels, and the flow is relatively low (low Reynolds number), the outlet temperature in many experiments approaches the surface temperature. Proper experiments need to be designed to account for this effect.

**(iii) Ambiguity (Karmaşa-LK) in the determination of the thermal boundary condition:** Several experiments reported in literature were conducted with microchannels or minichannels fabricated on **copper** or **silicon** substrates. The actual boundary conditions for these test sections are difficult to ascertain as they fall **in between** the **constant temperature** and **constant heat flux** boundary conditions. Furthermore, in many cases the heating is three-sided with the side walls acting as fins. The fin efficiency effects also alter the heat flux and temperature distributions. A clear comparison is only possible after the conjugate heat transfer effects are incorporated into a detailed numerical simulation of the test section.



**Fig. 3.17.** Selected experimental data for single-phase liquid flow in microchannels and minichannels,  $D_h = 50 \mu\text{m}$  to  $600 \mu\text{m}$ .

### 3.5.4. Roughness effects on laminar flow heat transfer

Kandlikar et al. (2003) studied the heat transfer and pressure drop of laminar flow in smooth and rough stainless steel tubes of 1.067 and 0.62 mm ID. The surface roughness of the inner tube wall was changed by treating it with two different acid mixtures. The surface roughness actually went down after the acid treatment as the protruding peaks in the surface profile were smoothed out. The effect of changes in the relative roughness on pressure drop was minimal, but the heat transfer in the entry region showed a distinct dependence on roughness. Figure 3.18 shows the local Nusselt number at a location 52 mm from the start of the heated length. The flow was hydrodynamically fully developed prior to entering the heated section.

It is seen from Fig. 3.18 that the Nusselt number increases even for a modest increase in the relative roughness from 0.161% to 0.355%. Also note that the location where the heat transfer measurements were taken was preceded by a tube length longer than the hydraulic entry length, which was 380 mm at  $Re = 2300$ . A number of researchers did not correctly identify the developing region and have mistakenly reported the increasing trend in the  $Nu$  versus  $Re$  plot as being in the fully developed region.

It is not possible to present any equations for heat transfer enhancement with roughness because of the lack of systematic data covering the entire region. Further research in this area is strongly recommended. With structured roughness surfaces, the enhancement results from the combined effects of area enhancement and that due to the flow modification caused by the roughness elements.

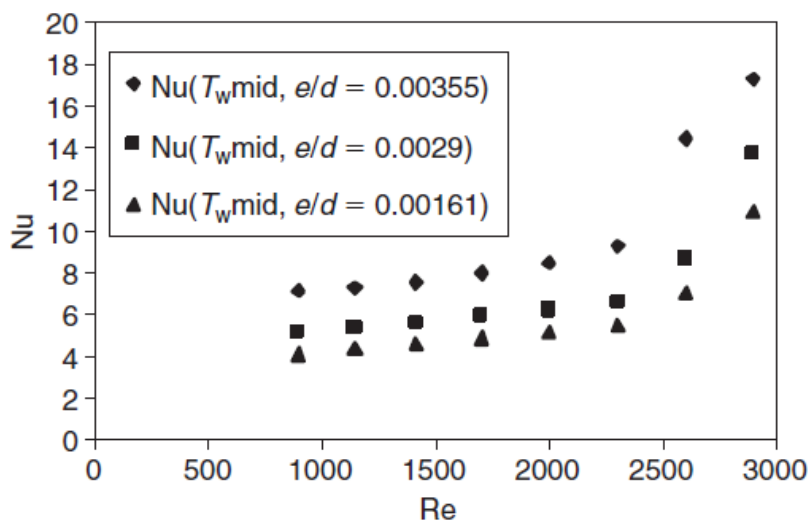


Fig. 3.18. The effect of roughness in the entrance region ( $x = 52$  mm) of a 0.62 mm inner diameter tube with a fully developed velocity profile and a developing temperature profile, Kandlikar et al. (2003).

### 3.5.5. Heat transfer in the transition and turbulent flow regions

A detailed discussion was presented earlier on the critical Reynolds number for the laminar-to-turbulent transition.

For smooth channels, the well-established criterion of  $Re_c = 2300$  is expected to hold. The effect of roughness is described by Eqs. (3.32) and (3.33). Since the transition region is encountered in many minichannel and microchannel heat exchangers, there is a need for generating accurate experimental data for both smooth and rough tubes in this region.

Phillips (1990) suggests using the following equations in the developing turbulent region. For larger values of  $x$ , the influence of the term  $[1 + (D_h/x)^{2/3}]$  reduces asymptotically to 1:

$$\text{For } 0.5 < Pr < 1.5: \quad Nu = 0.0214[1.0 + (D_h/x)^{2/3}][Re^{0.8} - 100]Pr^{0.4} \quad (3.47)$$

$$\text{For } 1.5 < Pr < 500: \quad Nu = 0.012[1.0 + (D_h/x)^{2/3}][Re^{0.87} - 280]Pr^{0.4} \quad (3.48)$$

Further validation of these equations in microchannels is warranted. Adams et al. (1997) conducted experimental work in the turbulent region with flow of water in 0.76 and 0.109 mm diameter circular channels. Based on their data, they proposed the following equation that also matches the data by Yu et al. (1995) within  $\pm 18.6\%$ :

$$Nu = Nu_{Gn}(1 + F) \quad (3.49)$$

where

$Nu_{Gn}$  represents the Nusselt number predicted by Gnielinski (1976) correlation.

$$Nu_{Gn} = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)} \quad (3.50)$$

$$f = (1.82 \log(Re) - 1.64)^{-2} \quad (3.51)$$

$$F = CRe(1 - (D/D_o)^2) \quad (3.52)$$

The least-squares fit to all the data sets studied by Adams et al. (1997) resulted in  $C = 7.6 \times 10^{-5}$  and  $D_o = 1.164$  mm.

Heat transfer coefficients in microchannels are very high due to their small hydraulic diameters. The high pressure gradients have led researchers to employ low flow rates.

However, with the reduced flow rate, the ability of the fluid stream to carry the heat away for a given temperature rise becomes limited. In order to improve the overall cooling performance, the following two options are available.

- (a) Reduce the flow length of the channels.
- (b) Increase the liquid flow rate.

As a result, employing multiple streams with short paths in a microchannel heat exchanger is recommended, similar to a split flow arrangement providing two streams as will be discussed in Section 3.7. The reduced flow length will then enable the designer to employ higher flow rates under a given pressure drop limit. This scheme offers several advantages over a single-pass arrangement where the fluid traverses the entire length of the heat exchanger:

(Advantages of employing multiple streams with short paths LK)

- (i) *The reduced flow length reduces the pressure drop*: The short flow length effectively reduces the overall pressure drop.
- (ii) *Larger developing region*: The multiple inlets result in a larger channel area being under developing conditions where the heat transfer is higher.
- (iii) *Higher flow velocities*: Some of the pressure drop reduction could be used to increase the flow velocity of individual streams. The possibility of employing turbulent flow should also be explored as the heat transfer coefficient is higher in this region.

### 3.5.6. Variable property effects

The **property ratio** method is usually recommended in accounting for the **property variations due to temperature changes** in the heat exchanger flow passages. The following equations are **recommended for liquids**:

$$f/f_{cp} = [\mu_w/\mu_b]^M \quad (3.53)$$

$$Nu/Nu_{cp} = [\mu_w/\mu_b]^N \quad (3.54)$$

where the subscript **cp** refers to the **constant property** solution obtained from appropriate equations or correlations. For **laminar flow**,  **$M = 0.58$**  and  **$N = -0.14$** , and for **turbulent flow**,  **$M = 0.25$**  and  **$N = -0.11$**  (Kays and London, 1984).

### 3.6. Microchannel and minichannel geometry optimization

The first practical implementation of microchannels in silicon devices was demonstrated by Tuckerman and Pease (1981). They were able to dissipate  $7.9 \text{ MW/m}^2$  with a maximum substrate temperature rise of  $71^\circ\text{C}$  and a pressure drop of  $186 \text{ kPa}$ .

The allowable temperature differences and the pressure drops for cooling today's microprocessor chips have become significantly smaller, and many investigators have been working towards optimization of the channel geometrical configuration.

The application of microchannels to electronics cooling imposes severe design constraints on the system design.

For a given heat dissipation rate, the

flow rate, pressure drop, fluid temperature rise, and

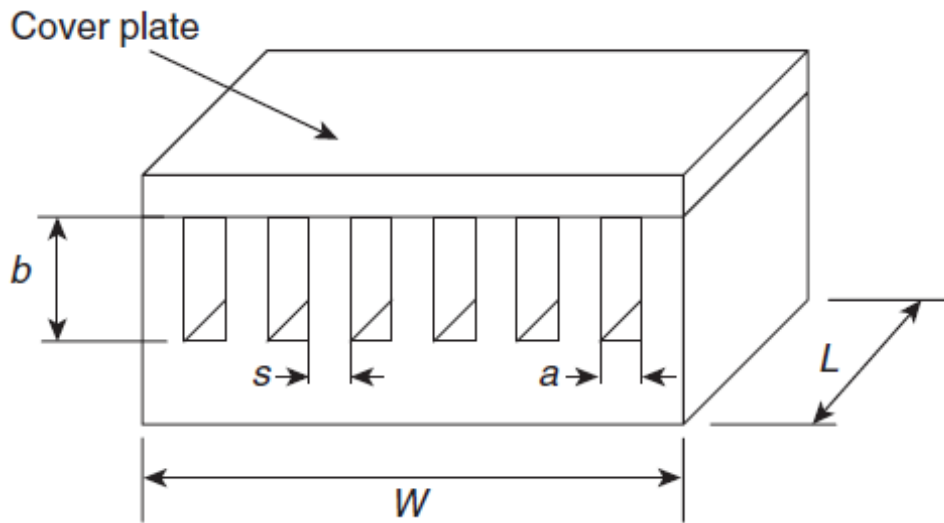
fluid inlet to surface temperature difference

requirements necessitate optimization of the channel geometry.

A number of investigators studied the geometrical optimization of microchannel heat exchangers. For example, Phillips (1987), Harpole and Eninger (1991), Knight et al. (1992), Ryu et al. (2002), Bergles et al. (2003), and Kandlikar and Upadhye (2005). The results found by Kandlikar and Upadhye are presented below.

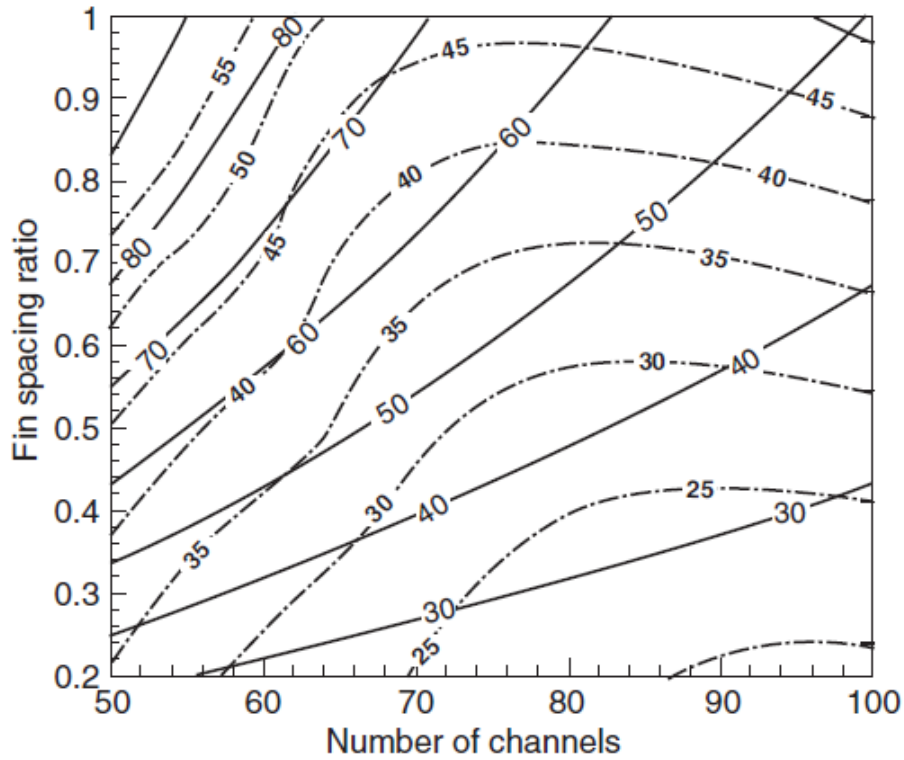
Kandlikar and Upadhye (2005) considered a microchannel system as shown in Fig. 3.19.

For a chip of width  $W = 10 \text{ mm}$  and length  $L = 10 \text{ mm}$ , they presented an analysis scheme for heat transfer and pressure drop by incorporating the entrance region effects. The number of channels was used as a parameter in developing the optimization scheme. The maximum chip temperature was set at  $360 \text{ K}$  while the fluid inlet temperature was set equal to  $300 \text{ K}$ . The channel depth was assumed to be  $200 \text{ }\mu\text{m}$ .

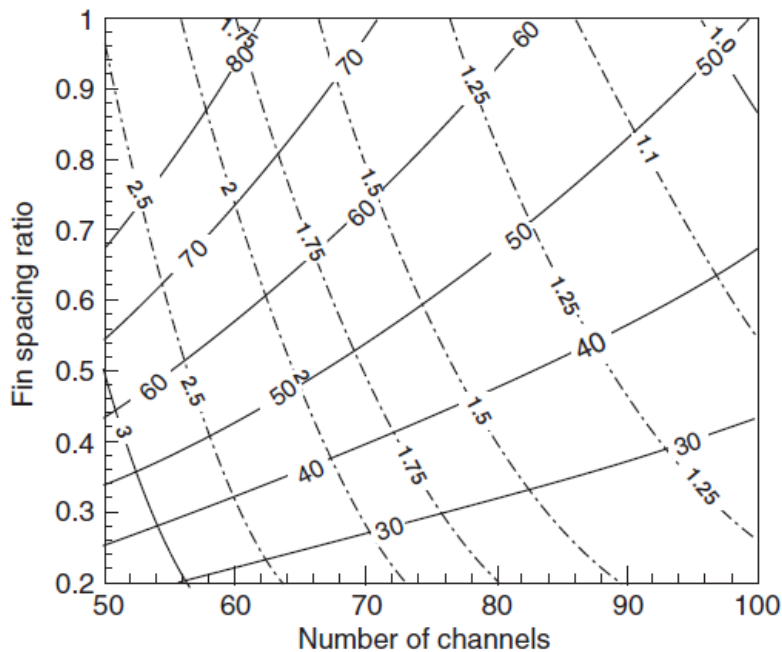


**Fig. 3.19.** Microchannel geometry used in microprocessor heat sink channel size optimization, Kandlikar and Upadhye (2005).

**Figures 3.20** and **3.21** show the parametric plots resulting from the **optimization program**. The fin spacing ratio, defined as  $\beta = s/a$ , is plotted as a function of the number of channels on a 10 mm wide chip with pressure drop, water flow rate and fin thickness as parameters. A lower limiting value of approximately 40  $\mu\text{m}$  was considered achievable for fin thickness using silicon microfabrication techniques. For a given pressure drop limit, the optimal number of channels and other channel geometrical parameters can be determined from this plot. Such plots could be generated for other system configurations as well.

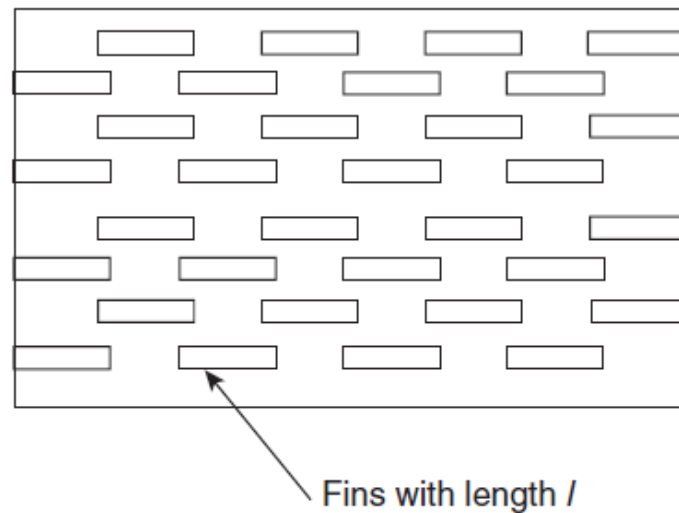


**Fig. 3.20.** Contour plot of fin spacing ratio  $\beta$  versus the number of channels with pressure drop across them in kPa (dash-dot lines) and fin thickness in  $\mu\text{m}$  (solid lines) as parameters for water flow in plain rectangular microchannels in a single-pass arrangement at a heat flux of  $3 \text{ MW/m}^2$ , Kandlikar and Upadhye (2005).

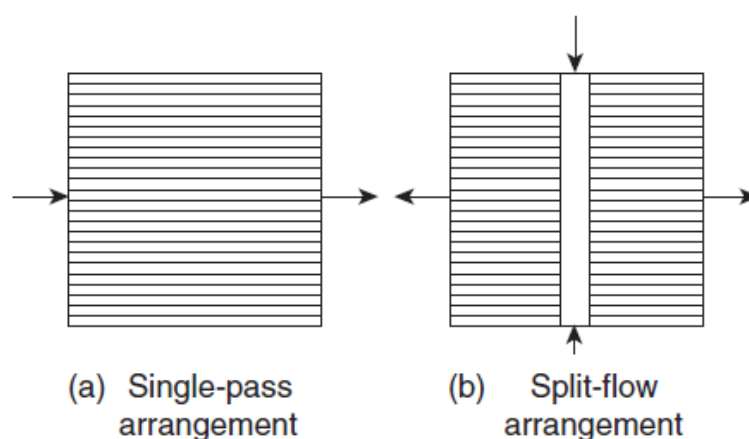


**Fig. 3.21.** Contour plot of fin spacing ratio  $\beta$  versus the number of channels with water flow rate in  $10^{-3} \text{ kg/s}$  (dash-dot lines) and fin thickness in  $\mu\text{m}$  (solid lines) as parameters for water flow in plain rectangular microchannels in a single-pass arrangement at a heat flux of  $3 \text{ MW/m}^2$ , Kandlikar and Upadhye (2005).

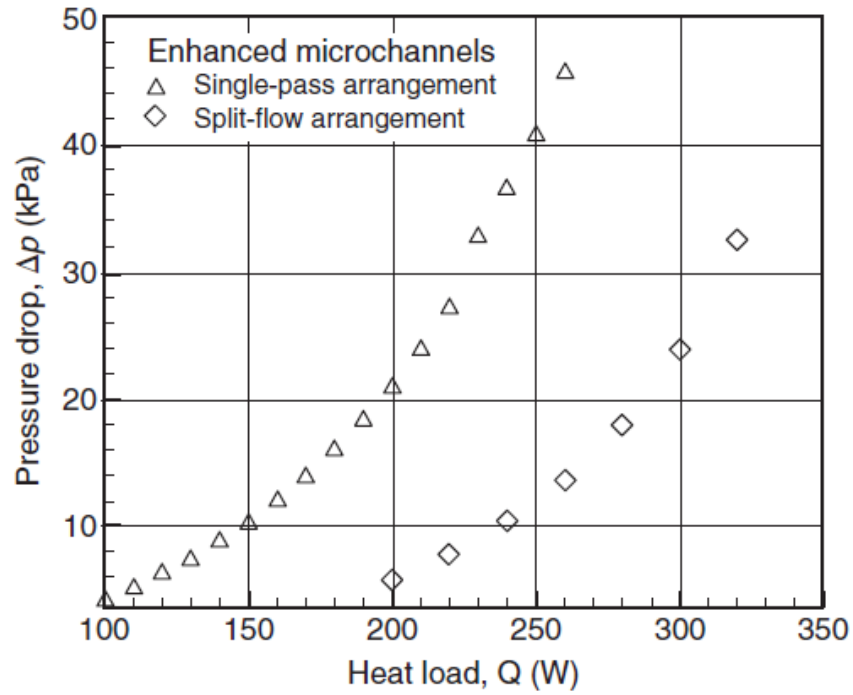
The effect of introducing offset strip-fins in the microchannel flow passages and a split flow arrangement, as shown in Figs. 3.22 and 3.23 respectively, was also analyzed by Kandlikar and Upadhye (2005). The resulting pressure drop versus the dissipated heat flux is shown in Fig. 3.24. It can be seen that the pressure drop is reduced considerably in the split flow arrangement.



**Fig. 3.22.** Offset strip-fins shown in the top view, with individual fin length  $l$  along the flow length, Kandlikar and Upadhye (2005).



**Fig. 3.23.** Schematics of single-pass and split-flow arrangements for fluid flow through microchannels, Kandlikar and Upadhye (2005).



**Fig. 3.24.** Comparison of pressure drops for the enhanced microchannels with offset strip-fins ( $l = 0.5$  mm) in single-pass and split-flow arrangements on a  $10 \text{ mm} \times 10 \text{ mm}$  chip, Kandlikar and Upadhye (2005).

### 3.7. Enhanced microchannels

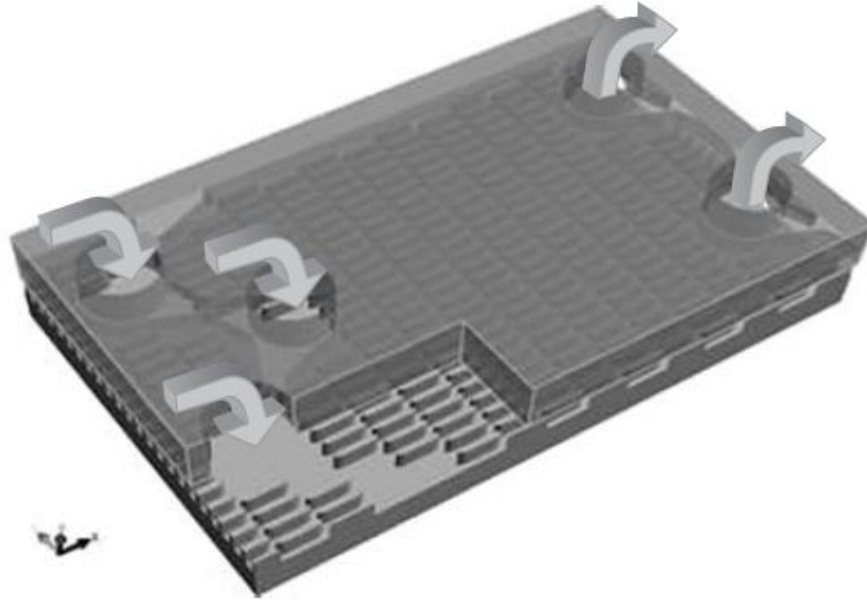
The use of enhanced microchannels was proposed by Kishimoto and Sasaki (1987).

The need for higher heat transfer coefficients than those attainable with plain microchannels was identified by Kandlikar and Grande (2004), and some specific enhancement geometries were suggested by Steinke and Kandlikar (2004). Kandlikar and Upadhye (2005) analyzed the enhanced offset strip-fin geometry and the results were presented in the previous section.

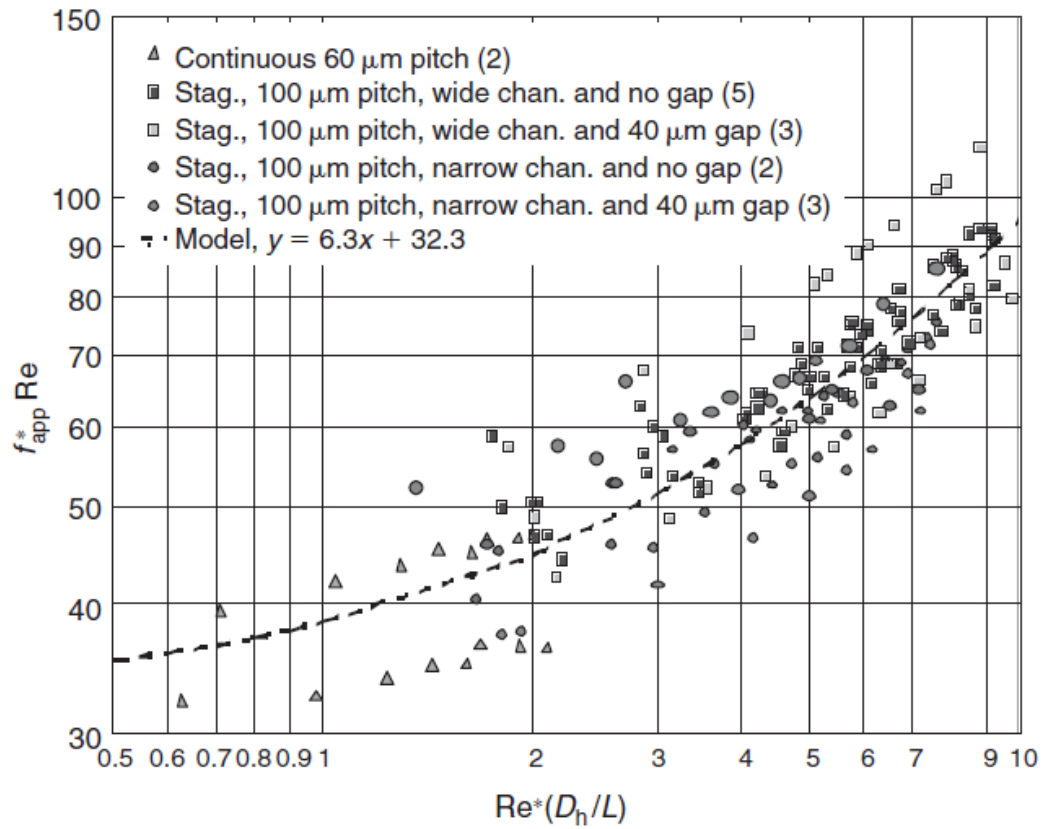
Colgan et al. (2005) presented detailed experimental results comparing various offset fin geometries. A three-dimensional rendition is shown in Fig. 3.25. There are several inlet and outlet manifolds, with short flow lengths through the enhanced structures.

The apparent friction factors and Nusselt numbers for plain and several enhanced configurations are shown in Figs. 3.26 and 3.27. A plot of  $f_{app}$  versus Stanton number is shown in Fig. 3.28. The relationship deviates somewhat from a linear behavior because the manifold effects are included in the data.

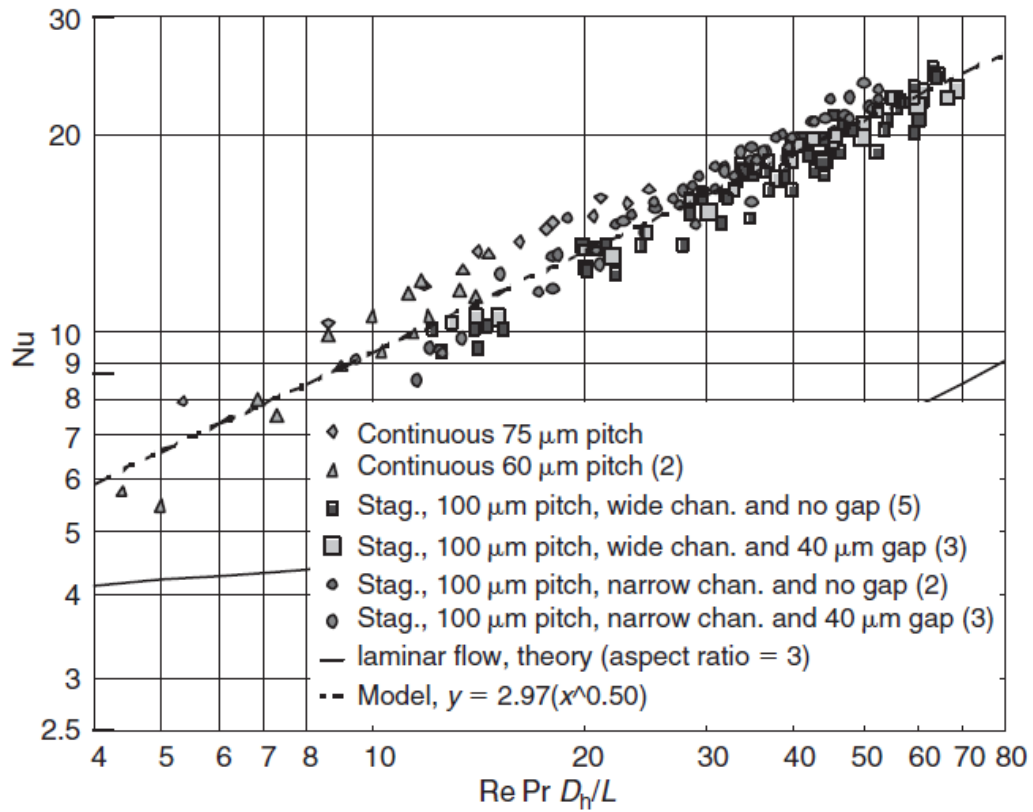
Microfabrication technology opens up a whole new set of possibilities for incorporating enhancement structures derived from past experience in compact heat exchanger development. Different manufacturing constraints enter into play for microchannels on a copper substrate as compared to the silicon microfabrication technology. Novel developments are expected to arise as we continue forward in integrating the microfabrication technology using creative ways to achieve high-performance heat transfer devices. Steinke and Kandlikar (2004) present a good overview of single-phase enhancement techniques and illustrate various enhancement configurations that may be considered especially with silicon microfabrication technology.



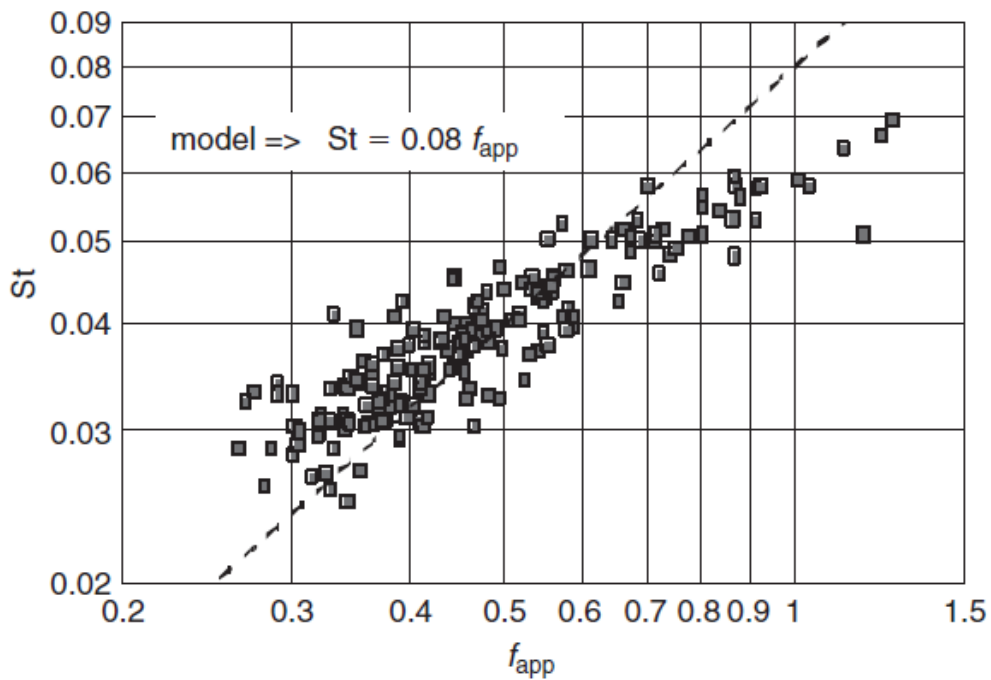
**Fig. 3.25.** Three-dimensional rendition of the IBM enhanced silicon chip with short multiple fluid streams. (Colgan et al., 2005).



**Fig. 3.26.** Apparent friction factor comparison of a plain and five enhanced microchannel configurations, Colgan et al. (2005).



**Fig. 3.27.** Nusselt number comparison of a plain and five enhanced microchannel configurations, Colgan et al. (2005).

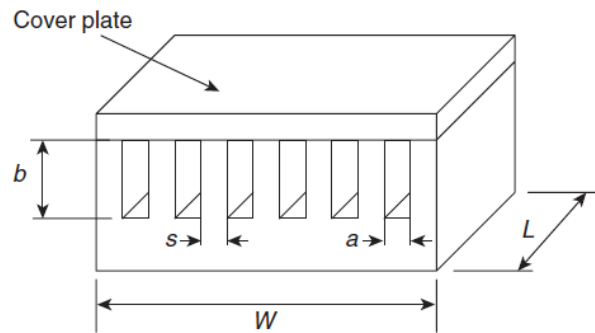


**Fig. 3.28.** Stanton number and friction coefficient relationship for microchannels with different fin geometries, Colgan et al. (2005).

## 3.8. Solved examples

### Example 3.1

Microchannels are directly etched into silicon in order to dissipate 100 Watts from a computer chip over an active surface area of  $10 \text{ mm} \times 10 \text{ mm}$ . The geometry may be assumed similar to Fig. 3.19. Each of the parallel microchannels has a width  $a = 50 \text{ }\mu\text{m}$ , depth  $b = 350 \text{ }\mu\text{m}$ , and a spacing  $s = 40 \text{ }\mu\text{m}$ . The silicon thermal conductivity may be assumed to be  $k = 180 \text{ W/m-K}$ .



**Fig. 3.19.** Microchannel geometry used in microprocessor heat sink channel size optimization, Kandlikar and Upadhye (2005).

Assume a uniform heat load over the chip base surface and a maximum water inlet temperature of  $35^\circ\text{C}$ . Assume one-dimensional steady state conduction in the chip substrate.

- (i) Calculate the number of flow channels available for cooling.
- (ii) Assuming the temperature rise of the water to be limited to  $10^\circ\text{C}$ , calculate the required mass flow rate of the water.
- (iii) Calculate the flow Reynolds number using the mean water temperature for fluid properties.
- (iv) Check whether the fully developed flow assumption is valid.
- (v) Calculate the average heat transfer coefficient in the channels.
- (vi) Calculate fin efficiency.
- (vii) Assuming the heat transfer coefficient to be uniform over the microchannel surface, calculate the surface temperature at the base of the fin at the fluid inlet and fluid outlet sections.
- (viii) Calculate the pressure drop in the core of the microchannel.
- (ix) If two large reservoirs are used as the inlet and outlet manifolds, calculate the total pressure drop between the inlet and the outlet manifolds.

### ***Solution***

*Properties:* Water at 40°C (Incropera and DeWitt (2002))

$\rho = 991.8 \text{ kg/m}^3$  ;  $\mu = 655 \times 10^{-6} \text{ N-s/m}^2$  ;  $c_p = 4179 \text{ J/kg-K}$ ;  $k = 0.632 \text{ W/m-K}$ ;

$\text{Pr} = 4.33$

**(i) Number of flow channels (Answer: 111 channels)**

**(i) Calculate the number of flow channels available for cooling.**

#### *Assumptions*

**(1)** The first and the last channels are half the channel width from the chip edge.

The remaining width left for channels is:  $10 \text{ mm} - 50 \mu\text{m} = 0.00995 \text{ m}$ .

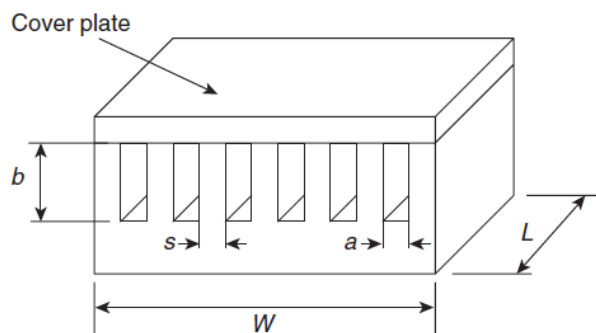
The pitch is:  $50 \mu\text{m} + 40 \mu\text{m} = 90 \mu\text{m} = 90 \times 10^{-6} \text{ m}$ .

Assumption 1 requires that the pitch repeat up to the last channel, but not include the last channel.

Therefore, the number of channels is found by:  $(0.00995 \text{ m} - 50 \times 10^{-6} \text{ m}) / (90 \times 10^{-6} \text{ m}) = 110$  channels.

But recall that that the above calculation was for a repeating pitch up to but not including the last channel so the number of channels available for cooling is

$n = 111$ .



**(ii) Required flow rate of water (Answer: 0.00239 kg/s or 143.6 ml/min, total)**

**(ii) Assuming the temperature rise of the water to be limited to 10°C, calculate the required mass flow rate of the water.**

The inlet water temperature is given as 35°C, and the stated assumption is that the water temperature rise is limited to 10°C. The mass flow rate can be calculated using the properties of water at the average temperature of 40°C by using the equation  $q = \dot{m}c_p\Delta T$  where  $q$  is the power dissipated from the chip,  $c_p$  is the specific heat of water at constant pressure and  $\Delta T$  is the temperature change of the water. The total mass flow rate is

$$\dot{m}_t = \frac{q}{c_p\Delta T} = 100 \text{ W} / (4179 \text{ J/kg-K})(10 \text{ K}) = 0.00239 \text{ kg/s} = 143.6 \text{ ml/min},$$

and the mass flow rate in one channel is given by

$$\dot{m}_c = \frac{q}{nc_p\Delta T} = 21.6 \times 10^{-6} \text{ kg/s} = 1.3 \text{ ml/min}$$

**(iii) Flow Reynolds number (Answer: Re = 164.7)**

**(iii) Calculate the flow Reynolds number using the mean water temperature for fluid properties.**

The flow Reynolds number using the mean water temperature (40°C) for fluid properties can be found using the hydraulic diameter  $D_h$  given by Eq. (3.7)

$$\begin{aligned} D_h &= \frac{2ab}{(a+b)} = 2(50 \times 10^{-6} \text{ m})(350 \times 10^{-6} \text{ m}) / (50 \times 10^{-6} \text{ m} + 350 \times 10^{-6} \text{ m}) \\ &= 87.5 \times 10^{-6} \text{ m} \end{aligned}$$

in the Reynolds number equation given by

$$\begin{aligned} \text{Re} &= \frac{\rho u_m D_h}{\mu} = \frac{\rho u_m A_c D_h}{A_c \mu} = \frac{\dot{m}_c D_h}{A_c \mu} \\ &= \frac{(21.6 \times 10^{-6} \text{ kg/s})(87.5 \times 10^{-6} \text{ m})}{(17.5 \times 10^{-9} \text{ m}^2)(655 \times 10^{-6} \text{ N-s/m}^2)} = 164.7 \end{aligned}$$

This low Reynolds number clearly indicates laminar flow regime.

**(iv) Fully developed assumption**

**(Answer: fully developed assumption is valid – hydrodynamic entrance length = 0.72 mm and thermal entrance length = 6.25 mm)**

**(iv) Check whether the fully developed flow assumption is valid.**

The fully developed flow assumption is valid if the thermal and hydrodynamic entrance lengths are less than the channel length. For laminar flow, the hydrodynamic entrance length is given by Eq. (3.11) as

$$L_h = 0.05 \text{ Re } D_h = 0.05(164.7)(87.5 \times 10^{-6} \text{ m}) = 720 \times 10^{-6} \text{ m} \\ = 0.72 \text{ mm},$$

and the thermal entrance length is given by Eq. (3.42) as

$$L_t = 0.1 \text{ Re Pr } D_h = 0.1(164.7)(4.33)(87.5 \times 10^{-6} \text{ m}) = 6.24 \times 10^{-3} \text{ m} \\ = 6.24 \text{ mm}$$

Since the

thermal and hydrodynamic entrance lengths are less than the channel length, the fully developed flow assumption is valid.

**(v) Average heat transfer coefficient (Answer:  $47.4 \times 10^3 \text{ W/m}^2 \cdot \text{K}$ )**

**(v) Calculate the average heat transfer coefficient in the channels.**

*Assumptions*

- (1) Constant heat flux boundary condition.
- (2) Three-sided heating condition.

The Nusselt number using the thermal conductivity of water at the average temperature (40°C) is given by  $\text{Nu} = h D_h / k$ . A value for the fully developed Nusselt number can be obtained from Table 3.3 using the aspect ratio of the channel  $\alpha_c = a/b = 1/7$ . By linear interpolation, the Nusselt number is  $\text{Nu}_{fd,3} = 6.567$ .

The average heat transfer coefficient is

$$\bar{h} = \frac{k \text{Nu}}{D_h} = (0.632 \text{ W/m-K})(6.567)/87.5 \times 10^{-6} \text{ m} = 47.4 \times 10^3 \text{ W/m}^2 \cdot \text{K}$$

**(vi) Fin efficiency (Answer: 67%)**

**(vi) Calculate fin efficiency.**

*Assumptions*

(1) Adiabatic tip condition.

For an adiabatic tip, the fin efficiency equation is given by  $\eta_f = \tanh(mb)/mb$  where  $b$  is the fin height

$(350 \times 10^{-6} \text{ m})$  and  $m$  is defined as  $m = (\bar{h}P/kA_c)^{1/2}$

In the above equation,  $k$  is thermal conductivity of the fin material (given as  $k = 180 \text{ W/m-K}$ ),  $A_c$  is the cross-sectional area of the fin  $L \times s = (0.01 \text{ m})(40 \times 10^{-6} \text{ m})$ , and  $P$  is the perimeter of the fin which can be defined as  $2L$  since the width of the fin is much smaller than its length. The term  $mb$  can be written as

$$\begin{aligned} mb &= \left( \frac{\bar{h}2L}{ksL} \right)^{1/2} b = \left( \frac{2\bar{h}}{ks} \right)^{1/2} b \\ &= \left( \frac{2(47.4 \times 10^3 \text{ W/m}^2\text{-K})}{(180 \text{ W/m-K})(40 \times 10^{-6} \text{ m})} \right)^{1/2} (350 \times 10^{-6} \text{ m}) = 1.270 \end{aligned}$$

Therefore, the fin efficiency can be calculated as

$$\eta_f = \tanh(1.270)/1.270 = 0.672 = 67\%.$$

**(vii) Inlet and outlet surface temperatures at the base of the fin**

**(Answer:  $T_{s,i} = 36.3^\circ\text{C}$  and  $T_{s,o} = 48.6^\circ\text{C}$ )**

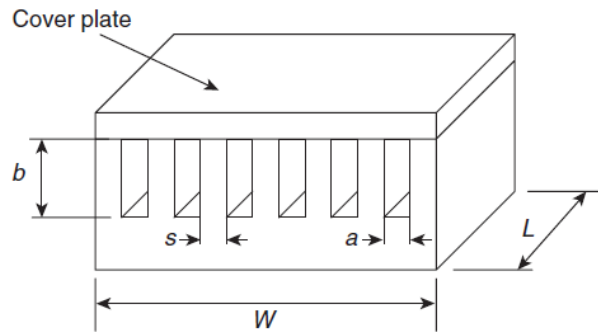
**(vii) Assuming the heat transfer coefficient to be uniform over the microchannel**

**surface, calculate the surface temperature at the base of the fin at the fluid**

**inlet and fluid outlet sections.**

*Assumptions*

(1) The heat flux is constant over the chip surface.



The surface heat flux considering the fin efficiency is given by

$$\begin{aligned}
 q'' &= q / (2b\eta_f + a)nL \\
 &= \frac{100 \text{ W}}{[2(350 \times 10^{-6} \text{ m})(0.672) + 50 \times 10^{-6} \text{ m}](111)(0.01 \text{ m})} \\
 &= 173 \times 10^3 \text{ W/m}^2
 \end{aligned}$$

The relationship between heat flux, heat transfer and temperature difference is given by  $q'' = h(T_s - T_f)$  where the subscripts s and f refer to the surface and fluid, respectively. The local heat transfer coefficients at the inlet and outlet of the microchannels are needed in order to compute the surface temperatures. Since the flow is developing at the entrance of the microchannel, Tables 3.3 and 3.4 are used with the channel aspect ratio and  $x^*$  for the inlet, where  $x^*$  is defined by Eq. (3.44). The entrance of the channel is assumed to begin at  $x = 0.1 \text{ mm}$  from the edge of the chip so  $x^*$  is

$$x^* = \frac{x/D_h}{\text{Re Pr}} \quad (3.44)$$

$$x_{\text{in}}^* = \frac{0.1 \times 10^{-3} \text{ m}}{(87.5 \times 10^{-6} \text{ m})(164.7)(4.33)} = 1.603 \times 10^{-3}$$

Using linear interpolation in Table 3.3, the fully developed Nusselt numbers for three-sided and four-sided heating are found to be 6.567 and 6.273, respectively. The thermal entry region Nusselt number for four-sided heating is found using Table 3.4 to be 18.42 for the inlet. Equation (3.45) is used to obtain the local Nusselt number at the entrance as

$$\text{Nu}_{x,3}(x^*, \alpha_c) = 18.42 \frac{6.567}{6.273} = 19.28$$

Alternatively, equations given in Appendix A may be used for interpolating values from Tables 3.2 to 3.4.

Using the thermal conductivity of water at the given inlet temperature (35°C), the local heat transfer coefficient at the inlet is found to be

$$h = \frac{k\text{Nu}}{D_h} = (0.625 \text{ W/m-K})(19.28)/87.5 \times 10^{-6} \text{ m} = 137.7 \times 10^3 \text{ W/m}^2\text{-K}$$

Since the flow is fully developed at the outlet, the three-sided Nusselt number from Table 3.3 is used with the thermal conductivity of water at the outlet temperature (45°C). The local heat transfer coefficient at the exit is found to be

$$h = \frac{k\text{Nu}}{D_h} = (0.638 \text{ W/m-K})(6.567)/87.5 \times 10^{-6} \text{ m} = 47.9 \times 10^3 \text{ W/m}^2\text{-K}$$

Using the value of the heat flux obtained above, the local heat transfer coefficients, and the given inlet and outlet water temperatures, the surface temperatures at the base of the fin at the fluid inlet and outlet are

$$T_s = \frac{q''}{h} + T_f \Rightarrow T_{s,i} = 36.3^\circ\text{C} \quad \text{and} \quad T_{s,o} = 48.6^\circ\text{C}$$

**(viii) Pressure drop in microchannel core (Answer: 43.6 kPa)**

**(viii) Calculate the pressure drop in the core of the microchannel.**

*Assumptions*

(1) The core of the microchannel includes pressure drop due to only frictional losses in the fully developed region and the loss due to the developing region.

Since the flow is fully developed at the exit, the pressure drop is defined by

$$\Delta p = \frac{2(f Re)\mu u_m L}{D_h^2} + K(\infty)\frac{\rho u_m^2}{2}$$

where  $K(\infty)$  is the Hagenbach factor, which is defined by Eq. (3.18) as

$$\begin{aligned} K(\infty) &= 0.6796 + 1.2197\alpha_c + 3.3089\alpha_c^2 - 9.5921\alpha_c^3 + 8.9089\alpha_c^4 - 2.9959\alpha_c^5 \\ &= 0.8969 \end{aligned}$$

The  $f Re$  term is given by Eq. (3.10) as

$$\begin{aligned} f Re &= 24(1 - 1.3553\alpha_c + 1.9467\alpha_c^2 - 1.7012\alpha_c^3 + 0.9564\alpha_c^4 - 0.2537\alpha_c^5) \\ &= 20.2 \end{aligned}$$

The pressure drop in the core of the microchannel is given by Eq. (3.14) as

$$\begin{aligned} \Delta p &= \frac{2(20.2)(655 \times 10^{-6} \text{ N-s/m}^2)(1.24 \text{ m/s})(0.01 \text{ m})}{(87.5 \times 10^{-6} \text{ m})^2} \\ &\quad + (0.8969)\frac{(991.8 \text{ kg/m}^3)(1.24 \text{ m/s})^2}{2} = 43.6 \text{ kPa} \end{aligned}$$

**(ix) Total pressure drop (Answer: 45.0 kPa)**

**(ix) If two large reservoirs are used as the inlet and outlet manifolds, calculate the total pressure drop between the inlet and the outlet manifolds.**

*Assumptions*

(1) Reservoirs are large so the Area of the reservoir  $\gg$  the Area of microchannel.

The total pressure drop between the inlet and outlet manifolds would include the pressure drop calculated above (43.6 kPa) plus the minor losses at the entrance and exit.

The minor loss is defined by  $\Delta p = K(\rho u_m^2/2)$  where  $K$  is a loss coefficient related to area changes at the entrance or exit. Taking the area ratio as zero, based on Assumption 1, for a laminar flow regime, the contraction and expansion loss coefficients  $K_c$  and  $K_e$  can be obtained from Fig. 3.5(b) as  $K_c = 0.8$  and  $K_e = 1.0$ .

The total pressure drop is obtained by

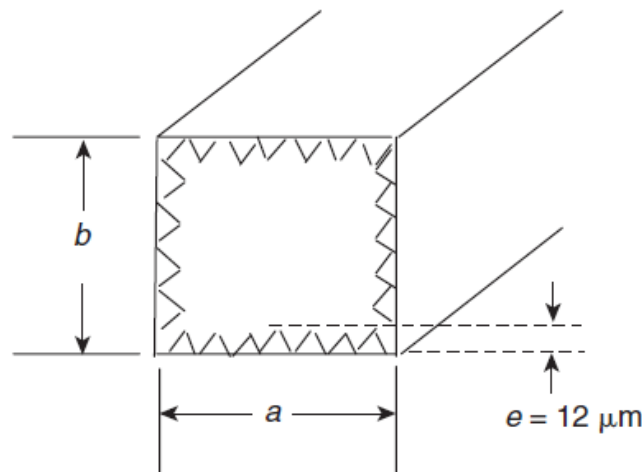
$$\Delta p = \frac{2(f \text{ Re})\mu u_m L}{D_h^2} + K(\infty)\frac{\rho u_m^2}{2} + K_c\frac{\rho u_m^2}{2} + K_e\frac{\rho u_m^2}{2} = 45.0 \text{ kPa}$$

The contraction and expansion coefficients were taken as the largest values on the chart in Fig. 3.5(b) in order to design for the maximum expected pressure drop.

### Example 3.2

A microchannel is etched in silicon. The microchannel surface is intentionally etched to provide an average roughness of  $12\ \mu\text{m}$ . The microchannel dimensions measured from the root of the roughness elements are: width –  $200\ \mu\text{m}$ , height –  $200\ \mu\text{m}$ , length –  $10\ \text{mm}$ . Water flows through the microchannels at a temperature of  $300\ \text{K}$ . Calculate the core frictional pressure drop when, (i)  $\dot{m} = 90 \times 10^{-6}\ \text{kg/s}$ , and (ii)  $\dot{m} = 180 \times 10^{-6}\ \text{kg/s}$ .

*Schematic*



*Solution*

*Properties:* From Incropera and DeWitt (2002), saturated water at  $300\ \text{K}$ :  
 $\mu_f = 0.855 \times 10^{-3}\ \text{N}\cdot\text{s}/\text{m}^2$ ,  $\rho = 997\ \text{kg}/\text{m}^3$ ,  $c_{p,f} = 4179\ \text{J}/\text{kg}\cdot\text{K}$ ,  $k_f = 0.613\ \text{W}/\text{m}\cdot\text{K}$ .

**(i) Core frictional pressure drop when  $\dot{m} = 90 \times 10^{-6}\ \text{kg/s}$  (Answer: 29,365 Pa)**

Calculate the hydraulic diameter using the constricted width ( $a_{cf}$ ) and constricted height ( $b_{cf}$ ) in Eq. (3.6)

$$D_{h,cf} = \frac{4A_c}{P_w} = a_{cf} = b_{cf} = 0.000176\ \text{m}$$

Calculate the Reynolds number of the fluid in a microchannel using Eq. (3.33)

$$Re_{cf} = \frac{\rho u_{m,cf} D_{h,cf}}{\mu} = \frac{(997.01\ \text{kg}/\text{m}^3)(2.91\ \text{m/s})(176 \times 10^{-6}\ \text{m})}{(0.855 \times 10^{-3}\ \text{Ns}/\text{m}^2)} = 598$$

For fully developed laminar flow, the hydrodynamic entry length may be obtained using Eq. (3.11)

$$L_{h,cf} = 0.05 \text{Re}_{cf} D_{h,cf} = 0.05(598)(0.176 \text{ mm}) = 5.26 \text{ mm}$$

Since  $L > L_{h,cf}$ , the fully developed flow assumption is valid.

The  $f \text{Re}$  term can be obtained using Eq. (3.10)

$$\begin{aligned} f \text{Re} &= 24(1 - 1.3553\alpha_c + 1.9467\alpha_c^2 - 1.7012\alpha_c^3 + 0.9564\alpha_c^4 - 0.2537\alpha_c^5) \\ &= 14.23 \end{aligned}$$

The core frictional pressure drop can be calculated using Eq. (3.14)

$$\Delta p = \frac{2(f \text{Re})\mu_{m,cf}L}{D_{h,cf}^2} + K(\infty)\frac{\rho u_{m,cf}^2}{2}$$

where  $K(\infty)$  is given by Eq. (3.18)

$$\begin{aligned} K(\infty) &= (0.6796 + 1.2197\alpha_c + 3.3089\alpha_c^2 - 9.5921\alpha_c^3 + 8.9089\alpha_c^4 - 2.9959\alpha_c^5) \\ &= 1.53 \end{aligned}$$

Hence

$$\begin{aligned} \Delta p &= \frac{2(14.23)(0.855 \times 10^{-3} \text{ Ns/m}^2)(2.91 \text{ m/s})(0.01 \text{ m})}{(176 \times 10^{-6} \text{ m})^2} \\ &\quad + (1.53)\frac{(997.01 \text{ kg/m}^3)(2.91 \text{ m/s})^2}{2} = 29,365 \text{ Pa} \end{aligned}$$

**(ii) Core frictional pressure drop when  $\dot{m} = 180 \times 10^{-6} \text{ kg/s}$**   
**(Answer: 68,694 Pa)**

Calculate the Reynolds number of the fluid in a microchannel using Eq. (3.33)

$$\text{Re}_{\text{cf}} = \frac{\rho u_{\text{m,cf}} D_{\text{h,cf}}}{\mu} = \frac{(997.01 \text{ kg/m}^3)(5.83 \text{ m/s})(176 \times 10^{-6} \text{ m})}{(0.855 \times 10^{-3} \text{ Ns/m}^2)} = 1200$$

For fully developed laminar flow, the hydrodynamic entry length may be obtained using Eq. (3.11)

$$L_{\text{h,cf}} = 0.05 \text{Re}_{\text{cf}} D_{\text{h,cf}} = 0.05(1200)(0.176 \text{ mm}) = 10.5 \text{ mm}$$

Since  $L < L_{\text{h,cf}}$ , the fully developed flow assumption **is not valid**.

Calculate the core frictional pressure drop with the developing laminar flow assumption. The apparent friction factor can be obtained from Table 3.2 using  $x^+$  calculated from Eq. 3.16

$$x^+ = \frac{x/D_{\text{h}}}{\text{Re}_{\text{cf}}} = \frac{0.01 \text{ m}}{(176 \times 10^{-6} \text{ m})(1200)} = 0.0475$$

From Table 3.2  $f_{\text{app}} \text{Re} = 21.35$  so the core frictional pressure drop can be calculated using the following equation

$$\begin{aligned} \Delta p &= \frac{2(f_{\text{app}} \text{Re}) \mu_{\text{m,cf}} L}{D_{\text{h,cf}}^2} \\ &= \frac{2(21.35)(0.855 \times 10^{-3} \text{ Ns/m}^2)(5.83 \text{ m/s})(0.01 \text{ m})}{(176 \times 10^{-6} \text{ m})^2} = 68,694 \text{ Pa} \end{aligned}$$

#### *Comments*

(1) The total frictional pressure drop has to consider the minor losses because the actual  $\Delta p$  is higher than the core frictional pressure drop.

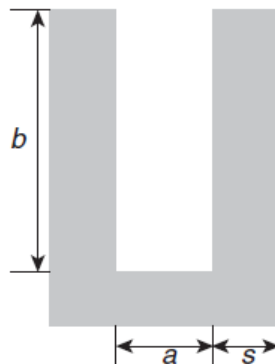
### Example 3.3

Consider a copper minichannel heat sink with an area of  $30 \text{ mm} \times 30 \text{ mm}$ , and relevant dimensions in Fig. 3.14 with  $a = 1 \text{ mm}$ ,  $b = 3 \text{ mm}$ , and  $s = 1.5 \text{ mm}$ . The heat dissipation is  $100 \text{ W}$  and the water inlet temperature is  $30^\circ\text{C}$ . The maximum surface temperature in the heat sink is limited to  $80^\circ\text{C}$ . Calculate the water flow rate under these conditions. Also calculate the frictional pressure drop in the core. Assume a constant heat transfer coefficient corresponding to fully developed conditions and take the thermal conductivity of copper to be  $400 \text{ W/m-K}$ .

#### Assumptions

Laminar flow, constant heat flux equally distributed over the surface area, one-dimensional steady state conduction, and three-sided heated condition with an adiabatic tip for fin efficiency

$a = 1 \text{ mm} = 10^{-3} \text{ m}$ ,  $b = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}$ ,  $s = 1.5 \text{ mm} = 1.5 \times 10^{-3} \text{ m}$ ,  
 $w = L = 30 \times 10^{-3} \text{ m}$ ,  $T_{f,i} = 30^\circ\text{C}$ ,  $T_{s,o} = 80^\circ\text{C}$ ,  $q = 100 \text{ W}$ ,  $k_{\text{Cu}} = 400 \text{ W/m-K}$ ,  
 $A_c = a \times b = 3 \times 10^{-6} \text{ m}^2$ ,  $\alpha_c = a/b = 1/3$



#### Solution

Equation 3.7 gives the hydraulic diameter,

$$D_h = \frac{2ab}{(a + b)} = \frac{2(10^{-3})(3 \times 10^{-3})}{(10^{-3} + 3 \times 10^{-3})} = 1.5 \times 10^{-3} \text{ m}$$

The Nusselt number can be obtained using Table 3.3 for a three-sided heated channel. Through interpolation this gives,

$$\text{Nu} = \text{Nu}_{\text{fd},3} = 5.224$$

For calculating water properties, let's assume the water exit temperature is equal to the maximum surface temperature of the heat sink. Then, the average water temperature is,

$$T_{\text{avg}} = (30^{\circ}\text{C} + 80^{\circ}\text{C})/2 = 55^{\circ}\text{C}$$

Properties of water at 55°C (from Incropera and DeWitt, 2002):

$$\rho = 985 \text{ kg/m}^3, \quad c_p = 4.183 \text{ kJ/kg-K}, \quad \mu = 505 \times 10^{-6} \text{ kg/m-s}, \\ k = 648 \times 10^{-3} \text{ W/m-K}$$

The average heat transfer coefficient is calculated using the Nusselt number which is rearranged to give,

$$\bar{h} = \frac{k\text{Nu}}{D_h} = \frac{(0.648)(5.224)}{(1.5 \times 10^{-3})} = 2.26 \times 10^3 \text{ W/m}^2\text{-K}$$

By relating the conductive and convective heat transfer for a constant heat flux, the average temperature difference between the surface and the fluid is given as

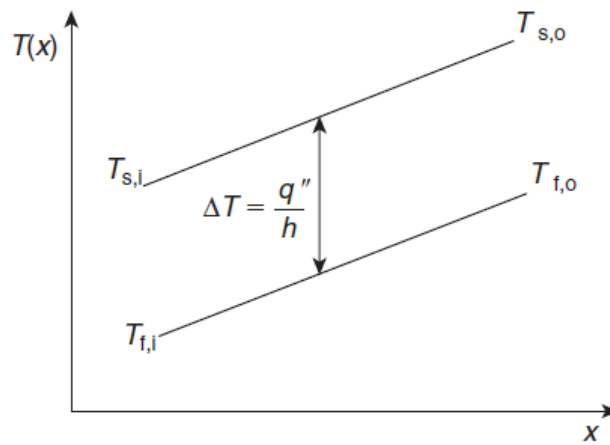
$$\Delta T = \frac{q}{h(2b\eta_f + a)nL}$$

The number of channels is,  $n = \frac{\text{overall width}}{\text{channel} + \text{fin}} = \frac{w}{a + s} = \frac{0.030}{0.001 + 0.0015} = 12$

The fin efficiency is,  $\eta_f = \frac{\tanh(mb)}{mb}$

$$mb = \sqrt{\frac{2\bar{h}}{k_{\text{Cu}}s}} \quad b = \sqrt{\frac{2(2.26 \times 10^3)}{(400)(1.5 \times 10^{-3})}} \times (3 \times 10^{-3}) = 0.260$$

$$\eta_f = \frac{\tanh(0.260)}{(0.260)} = 0.978$$



$$\Delta T = \frac{100}{(2.26 \times 10^3)[2(3 \times 10^{-3})(0.978) + 10^{-3}](12)(30 \times 10^{-3})} = 17.9^\circ\text{C}$$

The fluid outlet temperature is,  $T_{f,o} = T_{s,o} - \Delta T = 80^\circ\text{C} - 17.9^\circ\text{C} = 62.1^\circ\text{C}$

The average water temperature is,  $T_{\text{avg}} = (30^\circ\text{C} + 62.1^\circ\text{C})/2 = 46.1^\circ\text{C}$

$T_{\text{avg}}$  is off by 9 degrees. It is necessary to iterate again with the updated average temperature of  $46^\circ\text{C}$

Properties of water at  $46^\circ\text{C}$  (from Incropera and De Witt, 2002):

$\rho = 990 \text{ kg/m}^3$ ,  $c_p = 4.180 \text{ kJ/kg-K}$ ,  $\mu = 588 \times 10^{-6} \text{ kg/m-s}$ ,  $k = 639 \times 10^{-3} \text{ W/m-K}$

Following identical steps the new values are,

$$\bar{h} = 2.23 \times 10^3 \text{ W/m}^2\text{-K}, \quad mb = 0.258, \quad \eta_f = 0.978, \quad \Delta T = 18.2^\circ\text{C}, \quad T_{f,o} = 61.8^\circ\text{C}$$

$$T_{\text{avg}} = (30 + 61.8)/2 = 45.9^\circ\text{C}$$

The calculated average temperature is approximately the same as the assumed value. No more iterations are necessary.

$$T_{s,i} = T_{f,i} + \Delta T = 30^\circ\text{C} + 18.2^\circ\text{C} = 48.2^\circ\text{C}$$

$$q = \dot{m}c_p\Delta T_{\text{fluid}}$$

$$\dot{m}_t = \frac{q}{c_p\Delta T_{\text{fluid}}} = \frac{(100)}{(4180)(61.8 - 30)} = 752 \times 10^{-6} \text{ kg/s}$$

$$Q_c = \frac{\dot{m}_t(1/n)}{\rho} = \frac{(752 \times 10^{-6})(1/12)}{(990)} = 63.3 \times 10^{-9} \text{ m}^3/\text{s} \quad \text{or} \quad 3.80 \text{ ml/min}$$

$$u_m = \frac{Q_c}{A_c} = \frac{(63.3 \times 10^{-9})}{(3 \times 10^{-6})} = 0.0211 \text{ m/s}$$

$$\text{Re} = \frac{D_h u_m \rho}{\mu} = \frac{(1.5 \times 10^{-3})(0.0211)(990)}{(588 \times 10^{-6})} = 53.3$$

Flow is laminar so original assumption is correct.

$$\text{Equation 3.14 gives the pressure drop, } \Delta p = \frac{2(f\text{Re})\mu u_m L}{D_h^2} + K(\infty)\frac{\rho u_m^2}{2}$$

$f\text{Re}$  can be determined with Eq. 3.10,

$$f\text{Re} = 24(1 - 1.3553\alpha_c + 1.9467\alpha_c^2 - 1.7012\alpha_c^3 + 0.9564\alpha_c^4 - 0.2537\alpha_c^5) = 17.1$$

Hagenbach's factor comes from Eq. (3.22),

$$\begin{aligned} K(\infty) &= 0.6796 - 1.2197\alpha_c + 3.3089\alpha_c^2 - 9.5921\alpha_c^3 + 8.9089\alpha_c^4 - 2.9959\alpha_c^5 \\ &= 1.20 \end{aligned}$$

The pressure drop in the core is therefore,

$$\Delta p = \frac{2(17.1)(588 \times 10^{-6})(0.0211)(30 \times 10^{-3})}{(1.5 \times 10^{-3})^2} + (1.20) \frac{(990)(0.0211)^2}{2} = 5.92 \text{ Pa}$$

We need to calculate the hydrodynamic and thermal entrance lengths to justify fully developed conditions.

Using Eq. (3.11) we have,

$$L_h = 0.05 D_h \text{ Re} = 0.05(1.5 \times 10^{-3})(53.3) = 0.00400 \text{ m} = 4.00 \text{ mm}$$

Similarly Eq. (3.42) gives,

$$L_t = 0.05 D_h \text{ Re Pr} = 0.05(1.5 \times 10^{-3})(53.3)(3.85) = 0.0154 \text{ m} = 15.4 \text{ mm}$$

This validates the fully developed assumption as given in the problem statement.

### *Comments*

If the heat dissipation is increased to 200 W the flow rate will be 18.4 ml/min. This increases the pressure drop to 38.8 Pa. This illustrates the performance hit for increased heat dissipation and the need to develop microchannel cooling under turbulent conditions where the heat transfer coefficient will be larger. For turbulent flow more heat can be dissipated with minimal performance loss in pressure drop.

### 3.9. Practice problems

**Problem 3.1** Solve Example 1 with the substrate as copper, with a thermal conductivity of 380 W/m-K.

**Problem 3.2** For Example 3.1, plot the outlet surface temperature and pressure drop as a function of the channel depth  $b$  over a range of 200 to 600  $\mu\text{m}$ .

**Problem 3.3** For Example 3.1, plot the outlet surface temperature and pressure drop as a function of the channel width  $a$  over a range of 50 to 200  $\mu\text{m}$ .

**Problem 3.4** Solve Example 3.1 with a split flow arrangement. Neglect the area reduction caused by the central manifold.

**Problem 3.5** Solve Example 3.1 with copper substrate and a split flow arrangement. Assume the channel width to be 100  $\mu\text{m}$  and a depth of 800  $\mu\text{m}$ . Neglect the area reduction caused by the central manifold.

**Problem 3.6** For Example 3.3 with copper, plot the outlet surface temperature and the pressure drop as a function of the channel depth over a range of 200 to 2000  $\mu\text{m}$ .

**Problem 3.7** In Example 3.3, redesign the heat sink if the design heat load increases to 500 W. You may have to rework the channel geometry and flow arrangements.

**Problem 3.8** Design a microchannel heat exchanger to dissipate 800 Watts from a copper heat sink of 20 mm  $\times$  20 mm heated surface area. The inlet water temperature is 40°C, and the maximum surface temperature in the heat sink is desired to be below 60°C. Check your channel dimensions from manufacturing standpoint (provide the manufacturing technique you will be implementing). Do not neglect the temperature drop occurring in the copper between the base of the channels and the bottom surface of the heat sink receiving heat. Show the details of the manifold design.

**Problem 3.9** A copper minichannel heat sink with an area of 30 mm  $\times$  30 mm, and relevant dimensions in Fig. 3.14 are  $a = 1$  mm,  $b = 3$  mm, and  $s = 1.5$  mm. Calculate the maximum heat dissipation possible with an inlet water temperature of 30°C and the maximum surface temperature in the heat sink is limited to 80°C. Calculate the water flow rate under these conditions. Also calculate the frictional pressure drop in the core under these conditions.

**Problem 3.10** Design a minichannel heat exchanger to dissipate 5 kW of heat from a copper plate with a footprint of 10 cm × 12 cm. The plate surface temperature should not exceed 60°C and the inlet design temperature for water is 35°C. Calculate the water flow rate, outlet water temperature, and core frictional pressure drop. Compare the performance of (a) straight once-through flow passages and (b) split flow passages.

**Problem 3.11** Design the heat exchanger for Problem 3.10 in aluminum, and compare its performance with a copper heat sink.

## Appendix A

**Table A.1**  
Curve-fit equations for Tables 3.2, 3.3, and 3.4

| Equations | Constants |          |          |          |                       |           |
|-----------|-----------|----------|----------|----------|-----------------------|-----------|
|           | <i>a</i>  | <i>b</i> | <i>c</i> | <i>d</i> | <i>e</i>              | <i>f</i>  |
| 1         | 141.97    | −7.0603  | 2603     | 1431.7   | 14364                 | −220.77   |
| 2         | 142.05    | −5.4166  | 1481     | 1067.8   | 13177                 | −108.52   |
| 3         | 142.1     | −7.3374  | 376.69   | 800.92   | 14010                 | −33.894   |
| 4         | 286.65    | 25.701   | 337.81   | 1091.5   | 26415                 | 8.4098    |
| 5         | 8.2321    | 2.0263   | 1.2771   | 0.29805  | 2.2389                | 0.0065322 |
| 6         | 8.2313    | 1.9349   | −2.295   | 0.92381  | 7.928                 | 0.0033937 |
| 7         | 36.736    | 2254     | 17559    | 66172    | 555480                | 1212.6    |
| 8         | 30.354    | 1875.4   | 13842    | 154970   | 783440                | −8015.1   |
| 9         | 31.297    | 2131.3   | 14867    | 144550   | 622440                | −13297    |
| 10        | 28.315    | 3049     | 27038    | 472520   | 1783300               | −35714    |
| 11        | 6.7702    | −3.1702  | 0.4187   | 2.1555   | $2.76 \times 10^{-6}$ | NA        |
| 12        | 9.1319    | −3.7531  | 0.48222  | 2.5622   | $5.16 \times 10^{-6}$ | NA        |

### Equation form:

$$\begin{aligned} \text{Equations 1–4: } y &= \frac{a + cx^{0.5} + ex}{1 + bx^{0.5} + dx + fx^{1.5}}; & \text{Equations 5–10: } y &= \frac{a + cx + ex^2}{1 + bx + dx^2 + fx^3} \\ \text{Equations 11–12: } y &= a + bx + c(\ln x)^2 + d \ln x + ex^{-1.5} \end{aligned}$$

### Equation variables

Laminar flow friction factor in entrance region of rectangular ducts, Table 3.2:

y-variable – friction factor

x-variable – Equation 1:  $\alpha_c = 1.0$

Equation 2:  $\alpha_c = 0.5$

Equation 3:  $\alpha_c = 0.2$

Equation 4:  $0.1 = \alpha_c = 10$

Fully developed Nusselt number for three-side heating, Table 3.3:

y-variable – Nusselt number for three-sided heating

x-variable – Equation 5:  $\alpha_c$

Fully developed Nusselt number for four-side heating, Table 3.3:

y-variable – Nusselt number for four-sided heating

x-variable – Equation 6:  $\alpha_c$

Thermal entry region Nusselt numbers for four-sided heating, Table 3.4:

y-variable – Nusselt number for four-sided heating in the entry region

x-variable – Equation 7:  $\alpha_c = 0.1$

Equation 8:  $\alpha_c = 0.25$

Equation 9:  $\alpha_c = 0.333$

Equation 10:  $\alpha_c = 0.5$

Equation 11:  $\alpha_c = 1.0$

Equation 12:  $\alpha_c = 10$